

Social Transmission Bias: Evidence from an Online Investor Platform*

Pengfei Sui[†] and Baolian Wang[‡]

May 8, 2024

Abstract

Using data from a Twitter-like investor social platform, we document evidence consistent with self-enhancing transmission bias. We find investors are more likely to post about their better-performing stocks. Their followers are more likely to buy the posted stocks than others, and postings' effect on follow-up purchases is related to postings' perceived credibility. The performance-postings relationship is stronger among more volatile stocks and the relationship between postings and follow-up purchases is stronger among stocks with higher recent returns, shedding light on the spread of high-variance and extrapolative strategies. We also document that the social network features influential nodes.

Keywords: social network; self-enhancing transmission bias; social trading platform

JEL Code: D91, G41

* The authors are grateful to Bing Han, Sung Kwan Lee, Guangli Lu, Rui Shen, Jinfei Sheng (discussant), Wei Xiong (discussant), Yongxiang Wang and seminar participants at the 2021 SME Research Conference of CUHK-Shenzhen, 2022 CICF, UBS Quant Series 2022, Shenzhen University, and Xi'an Jiaotong University for helpful comments. All errors are our own.

[†] School of Economics and Management, Chinese University of Hong Kong, Shenzhen. Email: psui@cuhk.edu.cn.

[‡] Warrington College of Business, University of Florida. Email: baolian.wang@warrington.ufl.edu.

1. Introduction

The literature has long acknowledged the effects of the social network on individual investors' belief formation and trading (Shiller, 1984; Shiller and Pound, 1989; Hirshleifer, 2020). In this paper, using a novel dataset from a Twitter-like investor social network, we study how information is transmitted along the social network and how social transmission affects investor trading and performance.

There are at least two substantial empirical challenges in establishing a causal effect of social interaction on investor trading decisions (Kuchler and Stroebel, 2021). First, it is typically necessary to observe both the trading activities and the social network structure within the same dataset. Many studies resort to coarse measures such as geographic proximity and school ties as proxies for social linkage when direct observation of the social network is unavailable. Second, estimating peer effects faces the twin identification challenges of selection and common shocks, commonly known as the “reflection problem” (Manski, 1993). Selection occurs when investors with similar unobserved characteristics select into peer groups. Common shocks occur when group members, by virtue of their association, experience group-level shocks.

Investor social trading platforms offer an almost ideal setting for examining the effects of social interaction on trading. The investor social network from which we obtain our data resembles Twitter. Registered users can follow other registered users and be followed, allowing us to observe the *directional* followed-follower network links. Additionally, we have access to all the postings made on the platform. Many users link the social network with their brokerage accounts, making their trading records publicly available. To address the identification problem, we take advantage of an event in which the website disabled the posting function. This event serves as an exogenous

shock to the communication cost between linked investors, impacting active posters more than inactive ones. By leveraging this event, we employ a difference-in-differences strategy to estimate the social effects.

Our data come from a leading Chinese investor social platform, covering the period from June 24, 2016, to June 10, 2020. The sample includes approximately 10,000 investors, with over 40,000 unique followed-follower relationships. In total, the investors conducted 2.3 million trades (1.3 million purchases and 1.0 million sales) and posted 1.3 million comments on their profile pages.

We report several findings. First, an investor is more likely to post about her stocks the better her portfolio has performed. Among all the stocks held, investors are more likely to post about the better-performing ones, even after controlling for portfolio performance. This finding is consistent with self-enhancing transmission bias: investors like to tell others about their successes rather than their failures.

Second, we find that when an investor makes posts, their followers are more inclined to buy the discussed stocks compared to their other stocks. Two tests suggest this relationship is causal. Firstly, we conducted a falsification test utilizing the directional network structure. Let's consider two investors, A and B, where B follows A, but A does not follow B. According to the network effect, A's posts should influence B's subsequent trading, but B's posts should not affect A's trading. On the other hand, the reflection problem suggests that a common shock impacts both their posts and trading. Thus, it predicts that A's posts would affect B's trading, and vice versa. Our results indicate that followers' posts do not impact the trading behavior of the investors they follow, supporting the network effect and inconsistent with the reflection problem. Secondly, we

conducted a difference-in-differences analysis mentioned above, which further supports the causal interpretation of the impact of posts on followers' trading.

Third, we find that followers react differently to different postings. They are more likely to purchase the discussed stocks if those stocks are held by the followed investor than if they are not, suggesting that actions speak louder than words. Followers are also more likely to purchase the discussed stocks if the followed investor has a higher recent portfolio performance and are more likely to do so if the postings are submitted via an iOS device rather than other types of devices, such as Android and Windows.⁴ These findings show that the perceived credibility of the postings affects followers' follow-up purchases. Interestingly, followers' follow-up purchases are not affected by the investors' holding-period returns of the discussed stocks. While this may suggest that followers can debias the self-enhancing transmission bias, it may also be due to the structure of the social trading platform. Followers on the platform do not easily observe investors' holding-period returns, but the past returns of a stock are easily observable. Consistently, we find that followers are more likely to purchase the discussed stocks if those stocks performed well in the recent past.⁵

Fourth, followers do not perform better with the stocks they purchase after a post (follow-up purchases) than they do with non-follow-up purchases, suggesting that the online investor social network does not provide followers any information advantage. The follow-up purchases we observe are consistent with an attention-based interpretation (Barber and Odean, 2008): the postings bring the discussed stocks into the followers' choice set and increase their purchases.

⁴ Bertrand and Kamenica (2018) show that, in the US, owning an iOS device is strongly associated with being in the top quartile of the income distribution. Berg, Burg, Gombovic, and Puri (2020) find that owning an iOS device is strongly correlated with lower credit default.

⁵ Followers' follow-up purchases are not affected by the sentiment of the postings either, perhaps because most postings are extreme and emotional and hence may not be taken as reliable.

Consistent with this, we find that postings do not change the followers' selling of the discussed stocks, perhaps because attention is not constrained for selling (Barber and Odean, 2008).

Our next set of results examines what types of stock are more prone to social transmission bias. We document that volatility strengthens the relationship between holding-period returns and postings, suggesting that successes with more volatile stocks are more worth showing off. Another important characteristic is recent stock returns. As discussed above, followers are more likely to purchase the discussed stocks if these stocks have recently performed well. Relatedly, a stock's recent return strengthens the relationship between holding-period returns and postings, suggesting that the recency effect plays a role in that relationship (Tversky and Kahneman, 1974; Malmendier, 2021).

The above two findings have implications for the model of Han, Hirshleifer, and Walden (2022). First, the model predicts that, given self-enhancing transmission and followers' failure to debias, investors are more likely to buy past winners. Consequently, high-variance strategies, which generate a larger upper tail of high returns that are heavily broadcasted, tend to spread through the investor population. In Han et al.'s model, neither the relationship between holding-period returns and postings nor the relationship between postings and follow-up purchases is a function of stock volatility or recent stock returns (after controlling for holding-period returns). Our findings suggest that the spread of extrapolative and high-variance strategies may be stronger than predicted by their model. Second, Han, Hirshleifer, and Walden (2022) assume that message receivers (corresponding to our followers) are more likely to be affected if the message senders (corresponding to followed investors in our setting) have a higher performance. Our finding suggests that message receivers can also be affected through an attention channel.

We then investigate the structure of the social network. The social network we study features influential nodes, where a small number of investors have most of the followers, while many investors have none at all. Additionally, investors with more followers tend to write more posts. On a per-posting basis, influential investors have a similar impact on each follower's follow-up purchase. However, due to their larger number of followers and the larger number of postings, their impact within the social network is much more significant compared to other investors. Overall, these patterns result in the disproportionate importance of influential investors in social transmission.

We contribute to the growing field of “social economics and finance,” a term popularized by Hirshleifer (2020) for the study of how social processes shape economic outcomes. Many studies report evidence of peer effects (Hong, Kubik, and Stein, 2004, 2005; Brown, Ivkovic, Smith, and Weisbenner, 2008; Shue, 2013; Heimer, 2016; Ouimet and Tate, 2020; see Kuchler and Stroebel (2021) for a review).⁶ Most relevant to our study is the underexplored subfield of social transmission bias, which refers to the systematic directional modification of ideas or signals as they pass from person to person (Hirshleifer, 2020). Hirshleifer (2020) discusses five types of social transmission bias, with self-enhancing transmission bias particularly relevant—the tendency of investors to discuss their trades with others more after experiencing stronger performance than

⁶ Various types of social networks have been examined, such as education networks (Cohen et al., 2008; Lerner and Malmendier, 2013; Shue, 2013), geographic neighbors (Hong, Kubik, and Stein, 2004, 2005; Pool, Stoffman, and Yonker, 2015; Huang, Hwang, and Lou, 2021), coworker networks (Duflo and Saez, 2002, 2003; Maturana and Nickerson, 2019; Ouimet and Tate, 2020), relatives (Li, 2014), and various online social networks (Bailey et al., 2018b; Heimer, 2016; Cookson, Engelberg, and Mullins, 2023; Chen and Hwang, 2021). There are two prominent interpretations of such peer correlations: individuals may use social networks as a source of information or peers' actions can directly enter individuals' utility functions, as with a fear of missing out or a desire to “keep up with the Joneses” (Bursztyn et al., 2014). Many of the above studies document that social networks facilitate valuable information transmission. Evidence supporting the utility channel also exists. Agarwal, Mikhed, and Scholnick (2020), Bertrand and Morse (2016), and Kuhn, Kooreman, Soetevent, and Kapteyan (2011) show that households' consumption is affected by neighbors' consumption. Frydman (2016) reports neural evidence consistent with relative wealth concern. Charles, Hurst, and Roussanov (2009), D'Acunto, Rossi, and Weber (2023), and Han, Hirshleifer, and Walden (2023) study how visibility bias affects peers' consumption and savings.

after weaker performance (Han, Hirshleifer, and Walden, 2022). Han, Hirshleifer, and Walden (2022) model that when agents prefer sharing success over failures, extrapolative and high-variance strategies tend to spread through the investor population.⁷

We are not the first to document this bias. Several studies have found investors more likely to discuss their investment successes than their investment failures, which is consistent with our own findings (Kaustia and Knupfer, 2012; Heimer and Simon, 2015; Escobar and Pedraza, 2019; Huang, Hwang, and Lou, 2021; Ammann and Schaub, 2021; Lane, Lim, and Uzzi, 2022). Kaustia and Knupfer (2012) report a positive relationship between new stock market participation in Finland and neighbors' recent stock returns. Heimer and Simon (2015) report that an investor's short-term returns increase the frequency with which they contact other investors. Escobar and Pedraza (2019) conduct a study based on random assignment of students across classrooms in a financial education initiative in Colombia, revealing that favorable peer returns on single-stock trades increase the stock market entry of inexperienced investors. Huang, Hwang, and Lou (2020) find that an investor buying a stock causes her neighbors to buy stocks in the same industry, particularly if her portfolio has recently performed well. Lane, Lim, and Uzzi (2022) find that professional investors report their high returns more than their low returns in instant messages. Ammann and Schaub (2021) observe that posting by investors on an online social trading platform increase when their portfolio performs well and that postings are associated with followers' trading. However, the authors' tests cannot address the reflection problem, and they primarily focus on portfolio performance and do not study how stock characteristics affect social transmission.

⁷ Heimer (2014) documents a positive correlation between social interaction and active investment, consistent with Han, Hirshleifer, and Walden (2022).

We extend this literature in several aspects. First, in our data, we have access to information on social links, communications (i.e., postings), their content, and the responses, which allows us to provide more granular and direct evidence of the effects of social interactions on investors' trading decisions. Many other studies either lack data on social links and have to rely on coarse proxies or do not see the communications and responses. Additionally, the disabling of the posting function on the website enables us to provide causal evidence. Our more detailed data enable us to explore questions that would not have been feasible otherwise. For instance, we can identify which stocks are broadcasted and test whether followers are more likely to purchase the discussed stocks if they are held by the followed investor than if they are not. The level of data granularity also allows us to examine how stock characteristics affect the strength of the social transmission bias, which goes beyond the focus of the majority of the previous studies that primarily analyze *portfolio*-level performance without considering individual-security level factors.⁸

Second, we find the social transmission bias is more pronounced for stocks with higher volatility and stronger performance. Additionally, our analysis reveals the presence of influential players within the social network who play a disproportionate role in social transmission. Incorporating these two characteristics into a social transmission bias model can yield new insights and implications. We conjecture that the observed pattern, where the social transmission bias is stronger for more volatile and better-performing stocks, can provide valuable insights into why investors tend to buy recent winners and engage in riskier investments. Furthermore, the existence

⁸ To the best of our knowledge, Lane, Lim, and Uzzi (2022) is the only other study to examine how individual security performance affects communications. They study professional day traders and focus on the effect of strong ties while we document evidence on how social transmission is related to stock characteristics. In addition, we address the reflection problem.

of influential players may help elucidate why aggregate outcomes often exhibit a degree of unpredictability and are prone to errors.⁹

Third, the social transmission literature, particularly regarding the examination of whether message receivers can debias the effects of selected broadcasting, faces challenges due to the reflection problem. Kuchler and Stroebe (2021) highlight two identification strategies: random peer group assignment (Shue, 2013) and random shocks to existing peer groups (Bailey et al., 2022). Our analysis, using the temporary disabling of the posting function, can be broadly categorized as an example of the second strategy. What sets our approach apart from these two strategies is our test on the potential impact of followers on those who are followed, serving as a natural falsification test for the reflection problem. Given the proliferation of online social trading platforms like Public and eToro, many of which feature *directional* linkages, we anticipate that our empirical strategy holds implications for other similar cases.

Our study underscores the increasing significance of social media in financial markets, as exemplified by the recent GameStop event (Pedersen, 2022). Many investors now rely on social media as a source of information and for facilitating investor coordination. Bailey et al. (2018a) and Bailey et al. (2018b) construct a social connectedness index based on friendship links on Facebook and have studied the effect of social networks on various domains such as the housing markets (Bailey et al., 2018a), international trade (Bailey et al., 2021), product adoption (Bailey et al., 2022), the spread of COVID-19 (Kuchler, Russel, and Stroebe, 2021), institutional investors' portfolio choice and firm valuation (Kuchler et al., 2022), and co-movement in firm fundamentals and returns (Peng et al., 2023). For instance, Bailey et al. (2018b) demonstrate that individuals

⁹ Hirshleifer (2020) terms such a feature *idiosyncrasy*.

whose far-away friends experienced higher recent house price growth become more optimistic about future local house price growth, buy larger houses, and pay more for a given house. Li, Shin, and Wang (2022) find that investors use social media apps to coordinate pump-and-dump schemes in the cryptocurrency market. Pedersen (2022) emphasizes the importance of thought leaders in social media-based investment; our finding of influential investors supports his model prediction.

Research has made use of data from investor social networks, notably from StockTwits and Seeking Alpha. Using StockTwits data, Cookson and Niessner (2020) study the determinants of investor disagreement; Cookson, Engelberg, and Mullins (2022) provide evidence of echo chamber effects on investor beliefs; and Cookson, Fos, and Niessner (2022) document that disagreement facilitates informed trading. Chen and Hwang (2021), using server log data from Seeking Alpha, document that investors are more likely to share articles that are conducive to impression management. However, none of the above studies examine investor trading because the websites do not provide such data.

2. Snowball and data

2.1 Background on Snowball Finance

We use data from Snowball Finance (Xueqiu.com; Snowball hereafter), one of the most popular social networks among Chinese investors. ~~Founded in 2010, it has 12 million monthly active users.~~¹⁰ To a large extent, its structure is very similar to Stocktwits.

¹⁰ See <https://xueqiu.com/operation/guide>.

Snowball has two unique features that enable our empirical analysis. First, like Twitter, it allows users to share information, post opinions, and interact with other users. Snowball users can also follow other users and be followed. We observe the list of users and, for each user, the list of followers. Second, users can link their brokerage accounts with Snowball and thus reveal their transactions and holdings. In May 2019, when we first crawled the data, 13,661 users had a linked brokerage account. Investors can only choose when to start the link but not what to share. Once linked, all of an investor's trades will be available on Snowball. The trades before the link are not revealed, avoiding "backfilling." Investors' trading data are authentic because they are directly from investors' brokerage companies.

Figure 1 provides sample profile pages of Snowball users. Panel A shows a user's profile as she would see it, logged in as herself. Besides some market information and her own portfolio composition, the page lists, at the bottom, the postings of the investors she follows. Postings are ranked from the newest to the oldest. The website will direct the investor to a followed investor's profile if she clicks the alias. Panel B shows a user's profile as another user, logged in as herself, would see it. It shows the number of followers, the number of followed users, location (if available), a self-portrait of investment style, and recent postings. If the user has a linked brokerage account, that appears at the bottom right. Panel C shows the linked portfolio page, which shows performance history and historical trading records—the data we collect.

Users may link their brokerage accounts for several reasons. Some benefits and website functions are only available if they do. For example, the trading records shown in Panel C of Figure 1 can only be observed by users who link their accounts. Through collaboration with the brokerage firm, Snowball helps the linked user participate in IPO subscriptions and enhance the return on their cash balance (i.e., by participating in reverse repurchase agreements or money market mutual

funds). Investors may also enjoy the feeling of interacting with others and of being influential. As we show later, a post induces more follow-up purchases if it discusses a stock an investor owns.

2.2 Snowball data

From Snowball, we collect the data on investor trading, investor network, and investor postings from June 24, 2016, to June 10, 2020. Initially, we collected the data in May 2019. In June 2020, we collected the postings and trading, but not the network data. We obtained another snapshot of the network in October 2020. Our main empirical analysis uses the postings and trading for the whole period and the network snapshot from May 2019. The results are very similar if we use the October 2020 snapshot.

The investor trading data we collect from Snowball are very similar to those used by Odean (1998) and Barber and Odean (2001). We create a user file, a trade file, and a file with all the posts.

At the user level, we collect information on user identification (a unique number Snowball assigns), the user's accounts (including delinked ones), an active/closed flag for each account (measured at the time we collected the data), the accounts' initiation and close dates, and user location if available. About a third of users report their location at either the province or prefecture-city level.¹¹

We create a trade file with information on investor identification, account identification, trade date, trade time, asset symbol, type of record (buy/sell, dividend payment, IPO subscription, etc.), and trade price. However, the data do not include the number of shares for each asset or the total value of an account. The data report the value of each asset as a percentage of the total

¹¹ Unfortunately, like other social media websites, Snowball does not provide much demographic information.

portfolio value; for each trade, we collect the percentages before and after the trade. In Panel C of Figure 1, for example, the graph shows three trades. The first was on May 9, 2019; the percentage of stock SH600000 changed from 0% to 34.58% at a price of ¥11.12. On June 25, 2019, there were two trades. The fraction of stock SH600000 decreased from 34.01% to 11.86% at the price of ¥11.73, then decreased from 11.9% to 0% at the price of ¥11.76. From these three trades, we know that this investor bought stock SH600000 on May 9, 2019, at the price of ¥11.12, and sold it on June 25, 2019, at the price of ¥11.74.¹²

Our Snowball data do not cover all the trades, creating concern about sample selection. Snowball limits the number of trades available to other users to 200. If the number of trades is more than 200, only the most recent 200 are visible. The majority of accounts do not reach this limit, and for them, we have the entire trading history. The truncation is more likely to affect users who trade very frequently. Our data, therefore, overweight trading in the period immediately before we collected our data. We evaluate how this truncation affects our results in Section 3.4.3.

In addition, Snowball users can delink their brokerage accounts at any time. For the delinked accounts, the website keeps the link initiation date, delink date, and portfolio performance for the linked period, but not the trading records. The fact that the trading history of the delinked accounts is unavailable to us may cause a selection problem. However, in Figure A1 and Table A1 of the Appendix, we discuss this issue and find that the closing of these accounts does not seem to correlate with performance.

¹² $[(34.01\% - 11.86\%)/34.01\% * 11.73 + 11.86\%/34.01\% * 11.76] = 11.7405$.

2.3 Other data

We obtain daily stock price, return, and volume data from the China Stock Market & Accounting Research Database (CSMAR). CSMAR also provides price-adjustment factors to account for the impact of events such as stock splits and dividend distributions. When needed, we use these factors to adjust stock prices from CSMAR and Snowball.

2.4 Summary statistics

Figure 2 displays the distribution of our investors by province, which is very similar to the geographic distribution of the investor population. For example, Beijing, Shanghai, Guangdong, Jiangsu, and Zhejiang are the top five provinces by the number of stock accounts on the two Chinese exchanges and are also the top five by the number of Snowball investors.¹³

Table 1 presents the summary statistics of the investor network and their postings. We focus on 13,661 investors with an active linked brokerage account. There are 3,665 investors who do not follow any other investors and do not have any followers. Since we study investor networks, we exclude those investors from the analysis. We group investors into 19 groups by the number of followers. Table 1 reports the number of investors in each group, the number of followed-follower linkages, and the number of postings in each group. Our final sample has 9,906 investors, 40,629 followed-follower linkages, and 96,173 postings.

The distribution of the number of followers is highly skewed. Fifty-seven investors have more than 100 followers (five investors have over 1,000 followers), while 58.18% have no

¹³ See the Shanghai Stock Exchange fact book (http://www.sse.com.cn/aboutus/publication/monthly/documents/c/sse_stat_monthly_200812.pdf) and the Shenzhen Stock Exchange fact book (<https://www.szse.cn/market/periodical/documents/P020180328480036830826.html>).

followers. The 57 investors with more than 100 followers have 44% of the total followed-follower linkages. Investors with more followers are also more active in posting. For example, 4.09% (i.e., 1 – 95.91%) of investors have more than 10 followers and they collectively write 27.62% (i.e., 1 – 72.38%) of all the postings. Such a distribution of investor linkages and postings is interesting. In Table 9, we examine the implications of the network structure on social transmission.

Table 2 reports additional summary statistics of the investors. In Table 1, when measuring the number of followers, we count only those with a linked brokerage account. If we also count followers without a linked account, the average number of followers increases from 4.101 to 6.980. The distribution continues to be highly skewed.

The average number of posts per year is 20.592, of which 5.586 are on stocks the posting investor holds. The average number of stocks in an investor's portfolio is 3.804. The average number of trades per year is 105.309 and the annual portfolio turnover is 12.865. In our sample period, the annual market turnover is 10.138; our investors trade significantly more frequently.

Panel B of Table 2 reports the correlations between investor characteristics. Number of followers is positively correlated with number of posts, suggesting that influential investors are more active on social media. Number of followers is positively correlated with number of stocks in one's portfolio, suggesting that influential investors hold larger portfolios. Number of followers is not strongly correlated with other portfolio characteristics.

3. Empirical results

Subsection 3.1 tests the relationship between holding-period returns and the likelihood of broadcasting (i.e., writing a post). Subsection 3.2 tests how an investor's posting affects her

followers' follow-up trading—both purchases and sales. Subsection 3.3 examines the effect of the characteristics of the postings on follow-up trading. Subsection 3.4 examines the effect of stock characteristics on follow-up trading. Subsection 3.5 examines how social transmission varies across investors with different numbers of followers.

3.1 *Holding-period returns and postings*

In this subsection, we test whether the likelihood that an investor posts about her stock is a function of her holding-period return on that stock. Our methodology follows the disposition effect literature (Feng and Seasholes, 2005; Ben-David and Hirshleifer, 2012; Barber and Odean, 2013; Frydman and Wang, 2020). Specifically, we use the transaction dataset to construct a holding sample at the investor-stock-day level. Investor-days on which an investor does not post are included in the analysis. For example, if an investor bought Kweichow Moutai stock on January 2, 2019, and held it until January 31, 2019, there will be 22 observations (22 trading days). We exclude positions for which we do not have information on the purchase price, which occurs mainly because investors bought their stocks before they linked their brokerage account with Snowball.

We estimate the effect of holding-period return on postings using the following model:

$$Posting_{i,s,t+1} = \alpha + \beta_1 HPR_{i,s,t} + \gamma' X_{i,s,t} + \varepsilon_{i,s,t+1} \quad (1)$$

The dependent variable $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day $t+1$, and 0 otherwise. $HPR_{i,s,t}$ is the return since purchase for investor i 's holding of stock s on day t . $X_{i,s,t}$ is a set of control variables. Following Ben-David and Hirshleifer (2012), we include three control variables: $Sqrt(Time\ owned)$ is the square root of the number of days since

purchase, $\text{Log}(\text{Purchase price})$ is the natural logarithm of the weighted-average purchase price, and *Volatility* is the stock volatility calculated using daily returns over the six months prior to purchase. We estimate Equation (1) using linear regression to avoid the incidental parameter problem of nonlinear regressions such as logistic or probit models with many fixed effects included (Neyman and Scott, 1948; Lancaster, 2000).

Table 3 reports the regression results. For presentation purposes, the reported coefficients are multiplied by 100. In Column (1), we do not include any control variables or fixed effects. The coefficient of holding-period return is 0.334 ($t = 4.66$), suggesting that a 100-percentage-point increase in holding-period return increases the likelihood of writing a post by 0.334 percentage points. The unconditional probability of writing a post is 0.498 percentage points. A 0.334 percentage-point increase is economically significant. In Column (2), we add the three control variables. In Column (3), we also add investor fixed effects, date fixed effects, and stock fixed effects. The coefficients of holding-period return from these two specifications are higher than that from Column (1), as is the statistical significance.

In Column (4), instead of adding investor, date, and stock fixed effects separately, we add the stock fixed effects and the investor-day interactive fixed effects. With the latter, we essentially compare stocks held by an investor on the same day. All the variations come from the different holding-period returns of stocks held by that investor on that day. Column (4) shows that the coefficient of holding-period returns is 0.128 ($t = 3.03$). Comparing Columns (3) and (4) reveals that adding the investor-day fixed effects reduces the coefficient of holding-period returns by about two-thirds; we may conjecture that this is because an investor's posting likelihood increases not only with individual stock performance but also with portfolio performance. With the investor-day fixed effects, the effect of portfolio performance is subsumed. Consistent with this conjecture, in

Column (5), we include portfolio return and the three sets of fixed effects separately. In this specification, portfolio return has a strong positive coefficient and the coefficient on holding-period return is 0.234. This finding is consistent with previous studies (e.g., Heimer and Simon, 2015). Relative to the estimate in Column (3), this is about a 40% reduction.¹⁴

Figure 3 presents a graphic nonparametric view of the relationship between holding-period return and posting. Specially, we define dummy variables that each represent a five-percent interval of holding-period return and then estimate the following model:

$$\begin{aligned} Posting_{i,s,t+1} = & \alpha + \beta_1 I_{\{HPR_{i,s,t} \in [-40\%, -35\%]\}} + \beta_2 I_{\{HPR_{i,s,t} \in [-35\%, -30\%]\}} \cdots + \beta_{17} I_{\{HPR_{i,s,t} \in [40\%, +\infty)\}} \\ & + \gamma' X_{i,s,t} + \varepsilon_{i,s,t+1} \end{aligned} \quad (2)$$

$I_{\{HPR_{i,s,t} \in [-40\%, -35\%]\}}$ equals 1 if the holding-period return is greater than or equal to -40% but lower than -35%, and 0 otherwise. Other dummy variables are similarly defined. The omitted base group is when the holding-period return is less than -40%. When estimating Equation (2), we include the control variables and the three sets of fixed effects, as in Column (3) of Table 3. Figure 3 makes clear another aspect of the results in Table 3; namely, that the likelihood of posting increases in a near-monotonic fashion as we move from the group with the lowest holding-period return to the group with the highest.

3.2 Postings and follow-up purchases

In the analysis, we focus on postings that mention at least one stock that the investor holds. If investor i posts on day t about stock s that she holds, we check whether her followers, in the next

¹⁴ In the analysis of the relationship between holding-period returns and postings, we include the investors without followers. In Table A2, we replicate Table 3 by excluding investors without followers and find very similar results.

few days, are more likely to buy that stock than other stocks that investor i holds. In our main empirical analysis, we focus on the purchases made in the next five trading days. We do not want too short a window because followers can be affected by postings written several days ago. Too long a window, however, will contain too much noise.

We implement the following model:

$$Purchase_{j(i),s,(t+1,t+5)} = \alpha + \beta_1 Posting_{i,s,t} + \gamma' X_{i,s,t} + \varepsilon_{j(i),s,(t+1,t+5)} \quad (3)$$

The unit of analysis is investor i –follower $j(i)$ –stock s –day t . We use $j(i)$ to indicate that for investor i 's postings, we only analyze her followers (as opposed to all investors). Investor-days without postings are not included in our sample. $Purchase_{j(i),s,(t+1,t+5)}$ equals 1 if follower $j(i)$ purchases stock s in the next five trading days following the posting by her followed investor i on day t , and 0 otherwise. $Posting_{i,s,t}$ equals 1 if investor i posts about stock s on day t , and 0 otherwise. $X_{i,s,t}$ is a set of control variables.

We use an example to illustrate how we create our sample. Suppose that, on day t , investor i holds three stocks—A, B, and C—and writes about A but not the other two. Suppose also that she has two followers: j_1 and j_2 . Then, in our analysis, we will have six observations: two followers on three stocks. If more than one stock is mentioned, we flag each as being posted. If investors' postings increase their followers' tendency to buy the posted stocks, we expect $\beta_1 > 0$.

The results in Table 4 show that β_1 is indeed positive and statistically significant. Column (1) includes no control variables or fixed effects. Column (2) includes the three control variables: $Sqrt(Time\ owned)$, $Log(Purchase\ price)$, and $Volatility$.¹⁵ Column (3) adds investor, date, and

¹⁵ In Section 3.5 (i.e., Tables 7 and 8), we examine several other post and stock characteristics, including HPR , and find β_1 continues to be significant.

stock fixed effects. Column (4) adds stock fixed effects and investor-date interactive fixed effects. Besides being statistically significant, the effect of postings is also economically sizable. For example, in Column (4), the most stringent specification, β_1 , is 0.198 ($t = 2.84$). It indicates that an investor's posting increases her followers' likelihood of buying the posted stocks in the next five days by 0.198 percentage points. The unconditional probability of buying any stock in a five-day period is 0.546 percentage points. A 0.198-percentage-point increase represents an economically sizable 34.20% increase.

3.3 Postings and follow-up sales

In this subsection, we examine whether an investor's postings affect her followers' selling decisions. The unit of analysis is investor i –follower $j(i)$ –stock s –day t . For any day t when investor i posts on a stock (which must be in her portfolio at the end of day t), we examine how such a posting affects follower j 's decision to sell this stock relative to other stocks in follower j 's portfolio. We only include followers who hold stock s on day t because, given short-sale constraints, only they have these stocks available for sale. Investor-days without postings are not included in our sample.

We implement a model similar to Model (3):

$$Sale_{j(i),s,(t+1,t+5)} = \alpha + \beta_1 Posting_{i,s,t} + \gamma' X_{i,s,t} + \varepsilon_{j(i),s,(t+1,t+5)} \quad (4)$$

The dependent variable $Sale_{j(i),s,(t+1,t+5)}$ is a dummy equal to 1 if follower $j(i)$ sells stock s in the five trading days following a post of his/her followed investor i , and 0 otherwise. $Posting_{i,s,t}$ is a dummy equal to 1 if investor i posts about stock s on day t , and 0 otherwise. The control variables are as defined in Table 4.

Table 5 reports the results. The first observation is that the number of observations is much smaller than that of the follow-up purchase analysis. This is largely expected because all the discussed stocks are available for purchase, but only in a small fraction of cases are they available for sale by the followers. Such a finding provides the first evidence that postings are likely to be much less important to follow-up sales than to follow-up purchases.

In Table 5, the coefficient of *Posting* is positive and significant in Columns (1) and (2). However, it becomes significantly smaller and loses statistical significance once the fixed effects are included in Columns (3) and (4). For instance, in Column (4), we have the follower-date fixed effects; within a follower-date, we compare whether the follower is more likely to sell stocks that are discussed by her followed investors than other stocks. The average selling propensity in this sample is 11.593 percentage points; a 0.328 percentage-point increase is economically small. The results show that investor postings do not significantly affect followers' selling propensity.

Such an asymmetric finding between follow-up purchases and follow-up sales is consistent with an attention-based interpretation (Barber and Odean, 2008). Investor postings are likely to increase followers' attention to the discussed stocks and hence increase their tendency to buy these stocks. However, given short-sale constraints, an investor can only sell the stocks she holds. Because a typical individual investor only holds a small number of stocks, her selling decision is not much affected by limited attention (Barber and Odean, 2008).

Given the asymmetric finding, the remaining tests focus on follow-up purchases.

3.4 Robustness

In the first two subsections, we conduct two tests to mitigate the potential reflection problem (Manski, 1993) for the analysis of the relationship between posting and follow-up

purchases. In the last subsection, we report robustness tests on different model specifications and on the sample truncation issue discussed in Section 2.2.

3.4.1 Falsification test by reversing network direction

In the first test, we take advantage of the fact that the network is directional. Such a network structure is similar to that of Twitter, where user B follows A, but A does not necessarily follow B. The social network effect predicts that A's postings will affect B's trading, but not the other way around if A does not follow B. If the relationship between postings and follow-up purchases is purely driven by the reflection problem, we should expect that B's postings will affect A's trading even if A does not follow B.

Table 6 reports the analysis of how followers' postings affect their followed investors' trading. The analysis is the same as in Table 4, except that we reverse the direction of the network. In 5.71% of the cases, the linkage between A and B is bidirectional; we remove those linkages from this analysis. Across the different specifications, the coefficient of *Posting* varies from 0.000 to 0.036 and its *t*-value is always less than 1. These results are consistent with the social network effect and inconsistent with the reflection problem.

3.4.2 A natural experiment

Snowball suspended the posting function from April 13, 2019, to April 19, 2019. Snowball claimed that the suspension was for system upgrading. One floated rumor was that a user wrote a post that leaked a monetary policy change that soon the People's Bank of China implemented and Snowball was required by the government to investigate the incident. Either way, this event serves as an exogenous shock to the communication cost between linked investors. This event was unexpected and attracted significant investor attention.

We design a difference-in-differences estimate. We expect that this event affected investors who were active in posting more than investors who were inactive. Investors who were inactive in posting were not much affected as they did not use the posting function either way, while investors who were active in posting could not post in the suspension period. If posting is indeed how followers are affected by their followed investors, we expect that disabling the posting function will reduce active investors' influence on their followers more than on inactive investors.

Panel A of Figure 4 reports the posting propensity of both the active and inactive posting investors. An investor's posting propensity is the average daily probability that she posts about a held stock, calculated as the number of such posts divided by the number of investor-stock-day observations, calculated over the 90-day period before the disabling (i.e., from January 14, 2019, to April 12, 2019). Active posting investors are those whose posting propensity is higher than 0.1. In this analysis, we have 2,387 investors; 0.1 is about the 82nd percentile. Panel A shows that the inactive investors are indeed inactive; their posting propensity is close to zero through the 25-week period around the disabling week. Active investors could not post in week 0, but their postings quickly rebounded in week 1. We also see a statistically significant increase in the posting propensity of inactive investors, although the magnitude is small. This increase is likely driven by the increased attention to postings in general caused by the disabling.

If investors influence their followers through postings, we expect that, when the posting function was disabled, investors' self-enhancing postings would have a smaller effect on their followers' purchases. In other words, in normal periods, followers are more likely to buy their followed investors' winning stocks than losing stocks. In the disabling week, such asymmetry will weaken, especially so if the followed investors used to be active in postings. We implement Model (5) to estimate the effects of disabling the posting function on active and inactive investors:

$Purchase_{j(i),s,(t+1,t+5)}$

$$= \alpha + \sum_{w=-12}^{12} \beta_w Gain_{i,s,t} * RelativeWeek_w + \sum_{w=-12}^{12} \gamma_w RelativeWeek_w + \varepsilon_{i,j,s,(t+1,t+5)} \quad (5)$$

The dependent variable $Purchase_{j(i),s,(t+1,t+5)}$ is a dummy equal to 1 if follower j buys stock s in the five trading days following a post of his/her followed investor i , and 0 otherwise. $Gain_{i,s,t}$ is a dummy equal to 1 if investor i 's holding-period return on stock s on day t is positive, and 0 otherwise. We use the $Gain$ dummy variable instead of the continuous holding-period return measure because the latter is noisier and we have a relatively small sample in this analysis. $RelativeWeek_w$ is a dummy equal to 1 if day t is in the week that is w weeks after the week in which the posting function was disabled.

The β_w coefficients measure the social transmission effect. Because an investor is more likely to broadcast her success (Table 3) and her followers are more likely to buy the broadcasted stocks than the others (Table 4), we expect β_w to be positive in a normal period and that disabling the posting function will decrease β_w more for active than for inactive investors.

Panel B of Figure 4 reports the results. The two solid lines plot β_w (multiplied by 100) and the dotted lines show the upper and lower bounds of the 95% confidence intervals. In most of the studied weeks, active investors have a higher β_w than inactive investors. This is expected because active investors are more likely to broadcast their successes. Consistent with our predictions, week 0 shows a significant drop in β_w for active investors. It increased, however, for inactive investors, perhaps due to the increased investor attention to the website; followers can check posts written before the disabling and check their followed investors' position even without seeing a post about it. The relative decrease in β_w of the active investors is evident. Relative to the pre-period, the

relative drop in β_w for active investors is also statistically significant at the 5% level. The relative decrease quickly narrowed in the next two weeks. By week 3, active investors' β_w was again higher than that of inactive investors.

Except week 0 and for a few weeks following, β_w moves in tandem for the two groups of investors, although not perfectly. In fact, except for weeks -4 and 12, weeks 0–2 are the only weeks in which β_w is lower for active than for inactive investors. Overall, these results show that the parallel trends assumption holds reasonably and they support the causal interpretation of our results.

3.4.3 Other robustness tests

We report some robustness tests in the Appendix. First, in Table A3, we define *Purchase* using a one-day window (Panel A) or a three-day window (Panel B). In both specifications, the coefficient of *Posting* is positive, consistent with our main specification. The results also show that the effects of postings decay: the effect on five-day follow-up purchases is less than five times the effect on one-day follow-up purchases.

Second, in Table A4, we conduct the analysis using the October 2020 network snapshot—rather than that of May 2019, used in Table 4—with similar results.

Third, as discussed in Section 2.2, Snowball only displays the most recent 200 trades for each investor. We do not anticipate that this truncation will bias the analysis of the relationship between holding-period return and posting. However, it may affect the estimation of the relationship between postings and follow-up purchases. For example, if an investor posts on day t and a follower trades in the next several days, but her trades are truncated, our analysis will result in an underestimation of the relationship between posting and follow-up purchases. To quantify the potential bias, we re-estimate Model (3) by removing the follower-days in which an investor's

trades are truncated. For example, if follower j 's trades are truncated and her earliest displayed trade was on day t , we include her in the analysis only after day t . Table A5 shows that the results in Table 4 are robust to this specification. We obtain similar statistical significance for the coefficients of *Posting*. Consistent with the above underestimation conjecture, the coefficients become slightly larger.

Fourth, in Table 4, the unit of analysis is investor i –follower $j(i)$ –stock s –day t . For an investor-stock-day, there can be multiple observations if the investor has multiple followers and these observations may not be independent. In Table A6, we conduct the follow-up purchase analysis at the investor-stock-day level. Specifically, for each investor-stock-day, in Panel A, we calculate the average value of *Purchase* and use it as the dependent variable. In Panel B, the dependent variable is a dummy equal to 1 if any follower buys, and 0 otherwise. Again, we find similar results.

3.5 Heterogeneity tests

Everything else the same, investors may find some successes more worth showing off than others. It is also possible that followers may find some postings more worth following. In this section, we examine how the social transmission effect varies across post and investor characteristics (Subsection 3.5.1) and stock characteristics (Subsection 3.5.2). For post characteristics, we examine how the relationship between posting and follow-up purchases varies across different types of posts. For stock characteristics, we examine how that relationship varies

across different types of stocks and also how the relationship between holding-period return and posting varies across different types of stocks.¹⁶

3.5.1 Post characteristics

We consider the following post characteristics: (a) the holding-period return of the stock discussed in the post, (b) the sentiment of the post, (c) the investor's past-six-months portfolio return, (d) the device used for posting, and (e) whether the posted stock is held by the investor or not. To conduct the test, we augment Model (3) by adding an interaction term between a characteristic and *Posting*. For each characteristic, we report two specifications: one with investor, date, and stock fixed effects and the other with stock and investor-date fixed effects.

We define a *PositiveSentiment* dummy that equals 1 if the post has more positive-sentiment words than negative-sentiment words, and 0 otherwise. To measure the sentiment of a word in financial contexts, we follow Loughran and McDonald (2011) by developing negative and positive word lists. Specifically, we read 5,000 posts and manually label the sentiment of all the words, resulting in lists of 150 positive words and 121 negative words. Using this measure, about 68.80% of the posts have positive sentiment, 7.49% are negative, and the rest are neutral.¹⁷

The first four columns of Table 7 show that neither the coefficient of *Posting*HPR* nor the coefficient of *Posting*PositiveSentiment* is statistically distinguishable from zero. The coefficients

¹⁶ In Table A7, we examine whether the relationship between holding-period return and posting and the relationship between posting and follow-up purchases are functions of past market returns. We find that neither relationship is strongly related to past market returns.

¹⁷ In Table A8, we find very similar results if we define sentiment in other ways. First, instead of relying on the word lists we developed, we use the word lists provided by HowNet, a highly influential linguistic knowledge base in the Chinese natural language processing community (Dong and Dong, 2003; Qi et al., 2019). Second, instead of defining a *PositiveSentiment* dummy, we define a *NegativeSentiment* dummy as the number of negative words divided by the total number of words, the number of positive words divided by the total number of words, or the difference between the numbers of positive and negative words divided by the total number of words.

of *HPR* and *PositiveSentiment* are also insignificant. These findings may indicate that followers can debias the self-enhancing transmission bias. But other interpretations are possible. Most of the postings are emotional and investors may not consider them reliable information sources. On the Snowball platform, investors' holding-period returns are not directly displayed on their user interface (Panel C of Figure 1). If they were directly displayed, they might have had a stronger effect. A stock's past returns, however, can be easily accessed on the stock's profile page. In the next subsection, we test whether that has any effect.

While the holding-period return is not displayed, the investor's overall portfolio return is displayed. Columns (5) and (6) show that the coefficient of *Posting*Past 6 month portfolio return* is significantly positive, suggesting that postings made by better-performing investors are more influential.¹⁸ In Columns (7) and (8), we test whether followers react differently to postings made via iOS devices and by other devices. About 5.56% of the posts are made via an iOS device. Owning an iOS device is positively correlated with income (Bertrand and Kamenica, 2018) and low credit risk (Berg, Burg, Gombovic, and Puri, 2020). Followers may perceive investors with an iOS device as more capable and postings made via iOS more reliable. Consistent with this conjecture, the coefficient of *Posting*iOS* is significantly positive.

As shown in Table 2, investors often talk about stocks that they do not hold. In the last two columns, we examine whether ownership of the posted stocks has any effect. We create a *HeldStock* dummy that equals 1 if the investor who writes the post holds the posted stock, and 0 otherwise. The sample in this test is different from that of the others. In other tests, we only consider investor-days if at least one held stock is discussed in a post. In this test, we include all

¹⁸ The results are robust if we consider past-one-year portfolio returns, as reported in Table A9.

the investor-days when at least one stock—held or not—is discussed in a post. For each such day, we check whether followers are more likely to purchase stocks that are (a) discussed but not held, (b) discussed and held, or (c) held but not discussed. The non-discussed but held stocks are the base case in our regression. The control variables being inapplicable to the non-held stocks, we omit them in the regressions. The *Posting* coefficient captures the difference between the discussed but non-held stocks and the base case, while the coefficient of *Posting*HeldStock* captures the difference between the discussed and held stocks and the discussed but non-held stocks. The results show that the coefficient of *Posting* is significantly negative in Column (9); postings about a non-held stock increase the follow-up purchases of held stocks more than of the posted but non-held stocks. The coefficient of *Posting*HeldStock* is strongly positive. Overall, these results show that actions speak louder than words.

3.5.2 Stock characteristics

Han, Hirshleifer, and Walden (2022) predict that extrapolative, high-variance, and high-skewness strategies are more likely than others to spread through the investor population. In their model, neither the relationship between holding-period return and posting nor the relationship between posting and follow-up purchases is a function of stock characteristics. In this subsection, we test whether the two relationships are functions of a stock's recent past returns, volatility, or skewness. It is reasonable to expect that investors may find it more exciting to broadcast the successes of some stocks than of others and that followers' receptiveness may also be a function of the posted stocks. For recent-past stock returns, we consider three windows: past one month, past three months, and past six months. Volatility and skewness are measured in the 125 trading days before the purchase.

Table 8 presents the results of these characteristics on the relationship between postings and holding-period return (Panel A) and between postings and follow-up purchases (Panel B). An interesting pattern emerges: all coefficients of the interaction terms are positive—although not always statistically significant, providing strong evidence that the strength of social transmission varies across stock characteristics.

Many of the findings are expected. A stock's recent return strengthens the relationship between postings and holding-period return (Columns (1)–(6)) of Panel A), suggesting that the recency effect affects investors' posting decisions. Stock volatility strengthens the relationship between holding-period returns and posting, suggesting that successes on risky stocks are felt to be more worth showing off.

Overall, the results in Table 8 show that the strength of social transmission is a function of stock characteristics. More importantly, the effects of these characteristics strengthen Han, Hirshleifer, and Walden's (2022) prediction that extrapolative, high-variance, and high-skewness strategies are more likely to spread because they are more likely to be associated with a higher holding-period return and therefore are more likely to be discussed. However, in the authors' model—conditional on a holding-period return—the strength of transmission is not a function of stock characteristics. Our findings show that the spread of these strategies may be stronger than Han et al.'s model predicts.

3.6 Influential investors

In Table 1, we observe that a few investors have a disproportionately high fraction of the network linkages and that they write more posts. Yet the intuition that there are influential nodes in the network may be premature for two reasons. First, the correlation between the number of

followers and the number of postings is not conditional on holding-period return. Second, it is unclear whether postings made by investors with many followers are equally persuasive compared with postings made by other investors. Suppose their followers understand that their followed investors write a lot of posts and perceive them to be of low quality or that these followers simply pay relatively less attention to each of those investors' postings. We may then expect that these investors are less influential than the numbers in Table 1 suggest. In Table 9, we examine whether the strength of social transmission is related to the number of one's followers.

We examine the effects of the number of followers in the same way that we examined stock characteristics in Table 8.¹⁹ In Panel A, in the analysis of the relationship between holding-period return and posting, the coefficient of $HPR * \text{Log}(1 + \text{Number of followers})$ is positive, although only statistically significant in Column (3). In Panel B, in the analysis of the relationship between postings and follow-up purchases, the coefficient of $\text{Posting} * \text{Log}(1 + \text{Number of followers})$ is never significantly different from zero in the four specifications.

In Table A11, we examine the effect of the number of followers using a different specification. Specifically, we sort all the investors into groups by the number of followers and define dummy variables indicating each group. We then interact either *HPR* or *Posting* with these dummy variables. This specification is more flexible, as it relaxes the linearity assumption in Table 9. Consistent with the results of Table 9, neither the relationship between holding-period returns and postings nor the relationship between postings and follow-up purchases is strongly dependent on the number of followers.

¹⁹ The results are almost identical if we define the number of followers to include those without linked brokerage accounts. See Table A10 for the results.

Overall, the results show that the social transmission effect is not strongly related to the number of followers. At least the effect does not weaken for investors with a larger number of followers. Combined with the observations in Table 1, it is evident that there are influential nodes in the social network we study. Such a finding is consistent with Pedersen (2022), who emphasizes the importance of thought leaders in understanding social network effects on the financial markets.

3.7 Performance

Table 10 examines whether follow-up purchases are beneficial to the followers. If investors' postings contain value-relevant information, follow-up purchases may be more profitable. For each purchase, we track its performance for four horizons, ranging from one to six months, and calculate the size-decile adjusted buy-and-hold return. We form the size-decile portfolios based on market capitalization at the end of the previous month. A stock's size-decile adjusted return is the difference between the buy-and-hold raw return and the return of the decile portfolio that the stock belongs to. Our results are similar if we adjust the raw returns by the market or by the Capital Asset Pricing Model.

We regress the decile-adjusted returns on a *Follow-up purchase* dummy that equals 1 if follower j bought stock s that one of her followed investors held and posted about in the past five trading days, and 0 otherwise. For each horizon, we consider three specifications. We include the investor fixed effects in the first specification, stock and date fixed effects in the second, and all three sets of fixed effects in the third. The results are consistent for all specifications and all the horizons. The coefficient of *Follow-up purchase* is never statistically significant, suggesting that the follow-up purchases are not more profitable than other purchases.

4. Conclusions

Using data from Snowball, a Twitter-like social platform for investors, we study how a social network affects trading. Many investors link their brokerage accounts with Snowball and make their brokerage trading public. We document evidence consistent with the self-enhancing transmission bias. Specifically, we find that investors are more likely to write posts about stocks with a higher holding-period return. Their postings increase their followers' propensity to purchase the discussed stocks and those followers are more likely to make follow-up purchases if the investor has a higher recent portfolio return or if the postings are made via an iOS device. The strength of follow-up purchases is also significantly greater if the discussed stocks are held by the investor, suggesting that actions speak louder than words.

We find that the relationship between holding-period returns and postings and the relationship between postings and follow-up purchases are both functions of stock characteristics such as recent past returns, volatility, and skewness. For example, the relationship between holding-period return and posting is stronger for more volatile stocks and stocks with a higher recent return. Conditional on a post about a stock, followers are more likely to buy it if the stock has a higher recent past return. Han, Hirshleifer, and Walden (2022) predict that extrapolative, high-variance, and high-skewness strategies are more likely to spread; our findings suggest that the spread may be stronger than predicted by their model.

References

- Agarwal, Su., V. Mikhed, and B. Scholnick, 2020, Peers' income and financial distress: Evidence from lottery winners and neighboring bankruptcies, *Review of Financial Studies* 33: 433-472.
- Ammann, M., and N. Schaub, 2021, Do individual investors trade on investment-related internet postings? *Management Science* 67: 5301-5967.
- Bailey, M., R. Cao, T. Kuchler, J. Stroebel, and A. Wong, 2018a, Social connectedness: Measurement, determinants, and effects, *Journal of Economic Perspectives*, 32: 259-80.
- Bailey, M., R. Cao, T. Kuchler, and J. Stroebel, 2018b, The economic effects of social networks: Evidence from the housing market, *Journal of Political Economy* 126: 2224-2276.
- Bailey, M., E. Dávila, T. Kuchler, and J. Stroebel, 2019, House price beliefs and mortgage leverage choice, *Review of Economic Studies* 86: 2403-2452.
- Bailey, M., A. Gupta, S. Hillenbrand, T. Kuchler, R. Richmond, and J. Stroebel, 2021, International trade and social connectedness, *Journal of International Economics* 129: 103418.
- Bailey, M., D.M. Johnston, T. Kuchler, J. Stroebel, and A. Wong, 2022, Peer effects in product adoption, *American Economic Journal: Applied Economics* 14: 488-526.
- Barber, B., and T. Odean, 2001, Boys will be boys: Gender, overconfidence, and common stock investment, *Quarterly Journal of Economics* 116: 261-292.
- Barber, B., and T. Odean, 2008, All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors, *Review of Financial Studies* 21: 785-818.
- Barber, B., and T. Odean, 2013, The behavior of individual investors. In *Handbook of the Economics of Finance* (Vol. 2, pp. 1533-1570). Elsevier.
- Ben-David, I., and D. Hirshleifer, 2012, Are investors really reluctant to realize their losses? Trading responses to past returns and the disposition effect, *Review of Financial Studies* 25: 2485-2532.
- Berg, T., V. Burg, A. Gombovic, and M. Puri, 2020, On the rise of FinTechs: Credit scoring using digital footprints, *Review of Financial Studies* 33: 2845-2897.
- Bertrand, M., and E. Kamenica, 2018, Coming apart? Cultural distances in the United States over time, working paper.
- Bertrand, M., and A. Morse, 2016, Trickle-down consumption, *Review of Economics and Statistics* 98: 863-879.
- Brown, J., Z. Ivkovic, P. Smith, and S. Weisbenner, 2008, Neighbors matter: Causal community effects and stock market participation, *Journal of Finance* 63: 1509-1531.
- Bursztyn, L., F. Ederer, B. Ferman, and N. Yuchtman, 2014, Understanding mechanisms underlying peer effects: Evidence from a field experiment on financial decisions, *Econometrica* 82: 1273-1301.
- Charles, K., E. Hurst, and N. Roussanov, 2009, Conspicuous consumption and race, *Quarterly Journal of Economics* 124: 425-467.
- Chen, H., and B. Hwang, 2021, Listening in on investors' thoughts and conversations, *Journal of Financial Economics* forthcoming.
- Cohen, L., A. Frazzini, and C. Malloy, 2008, The small world of investing: Board connections and mutual fund returns, *Journal of Political Economy* 116: 951-979.
- Cookson, J. T., J. Engelberg, and W. Mullins, 2023, Echo chambers, *Review of Financial Studies* 36: 450-500.
- Cookson, J. T., S. Fos, and M. Niessner, 2022, Does disagreement facilitate informed trading? Evidence from activist investors, working paper.
- Cookson, J. T., and M. Niessner, 2020, Why don't we agree? Evidence from a social network of investors, *Journal of Finance* 75: 173-228.

- Dong, Z., and Q. Dong, 2003, HowNet-a hybrid language and knowledge resource. In *IEEE International Conference on Natural Language Processing and Knowledge Engineering Proceedings*: 820-824.
- Duflo, E., and E. Saez, 2002, Participation and investment decisions in a retirement plan: The influence of colleagues' choices, *Journal of Public Economics* 85: 121-148.
- Duflo, E., and E. Saez, 2003, The role of information and social interactions in retirement plan decisions: Evidence from a randomized experiment, *Quarterly Journal of Economics* 118: 815-842.
- D'Acunto, F., A. Rossi, and M. Weber, 2023, Crowdsourcing peer information to change spending behavior, *Journal of Financial Economics* forthcoming.
- Escobar, L., and A. Pedraza, 2019, Active trading and (poor) performance, working paper.
- Feng, L., and M. Seasholes, 2005, Do investor sophistication and trading experience eliminate behavioral biases in financial markets? *Review of Finance* 9: 305-351.
- Frydman, C., 2016, Relative wealth concerns in portfolio choice: Neural and behavioral evidence, working paper.
- Frydman, C., and B. Wang, 2020, The impact of salience on investor behavior: Evidence from a natural experiment, *Journal of Finance* 75: 229-276.
- Han, B., D. Hirshleifer, and J. Walden, 2022, Social transmission bias and investor behavior, *Journal of Financial and Quantitative Analysis* 57: 390-412.
- Han, B., D. Hirshleifer, and J. Walden, 2023, Visibility bias in the transmission of consumption beliefs and undersaving, *Journal of Finance* 78: 1647-1704.
- Heimer, R., 2014, Friends do let friends buy stocks actively, *Journal of Economic Behavior & Organization*, 107B: 527-540.
- Heimer, R., 2016, Peer pressure: Social interaction and the disposition effect, *Review of Financial Studies* 29: 3177-3209.
- Heimer, R., and D. Simon, 2015, Facebook finance: How social interaction propagates active investing, working paper.
- Hirshleifer, D., 2020, Presidential address: Social transmission bias in economics and finance, *Journal of Finance* 75: 1779-1831.
- Hong, H., J. Kubik, and J. Stein, 2004, Social interaction and stock-market participation, *Journal of Finance* 59: 137-163.
- Hong, H., J. Kubik, and J. Stein, 2005, Thy neighbor's portfolio: Word-of-mouth effects in the holdings and trades of money managers, *Journal of Finance* 60: 2801-2824.
- Huang, S., B. Hwang, and D. Lou, 2021, The rate of communication, *Journal of Financial Economics* 141: 533-550.
- Kaustia, M., and S. Knupfer, 2012, Peer performance and stock market entry, *Journal of Financial Economics* 104: 321-338.
- Kuchler, T., Y. Li, L. Peng, J. Stroebe, and D. Zhou, 2022, Social proximity to capital: Implications for investors and firms, *Review of Financial Studies* 35: 2743-2789.
- Kuchler, T., D. Russel, and J. Stroebe, 2021, JUE Insight: The geographic spread of COVID-19 correlates with the structure of social networks as measured by Facebook, *Journal of Urban Economics* forthcoming.
- Kuchler, T., and J. Stroebe, 2021, Social finance, *Annual Review of Financial Economics* 13: 37-55.
- Kuhn, P., P. Kooreman, A. Soetevent, and A. Kapteyn, 2011, The effects of lottery prizes on winners and their neighbors: Evidence from the Dutch postcode lottery, *American Economic Review* 101: 2226-2247.

- Lancaster, T., 2000, The incidental parameter problem since 1948, *Journal of Econometrics* 95: 391-413.
- Lane, J., S. Lim, and B. Uzzi, 2022, Biased information transmission in investor social networks: Evidence from professional traders, working paper.
- Lerner, J., and U. Malmendier, 2013, With a little help from my (random) friends: Success and failure in post-business school entrepreneurship, *Review of Financial Studies* 26: 2411-2452.
- Li, G., 2014, Information sharing and stock market participation: Evidence from extended families, *Review of Economics and Statistics* 96: 151-160.
- Li, T., D. Shin, and B. Wang, 2022, Cryptocurrency pump-and-dump schemes, working paper.
- Loughran, T., and B. McDonald, 2011, When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks, *Journal of Finance* 66: 35-65.
- Malmendier, U., 2021, Experience effects in finance: Foundations, applications, and future directions, *Review of Finance* 25: 1339-1363.
- Manski, C., 1993, Identification of endogenous social effects: The reflection problem, *Review of Economic Studies* 60: 531-542.
- Maturana, G., and J. Nickerson, 2019, Teachers teaching teachers: The role of workplace peer effects in financial decisions, *Review of Financial Studies* 32: 3920-3957.
- Neyman, J., and E. Scott, 1948, Consistent estimates based on partially consistent observations, *Econometrica* 16: 1-32.
- Odean, T., 1998, Are investors reluctant to realize their losses? *Journal of Finance* 53: 1775-1798.
- Ouimet, P., and G. Tate, 2020, Learning from coworkers: Peer effects on individual investment decisions, *Journal of Finance* 75: 133-172.
- Pedersen, L., 2022, Game on: Social networks and markets, *Journal of Financial Economics* 146: 1097-1119.
- Peng, L., S. Titman, M. Yonac, and D. Zhou, 2023, It's a small world: Social ties, comovements, and predictable returns, working paper.
- Pool, V.K., N. Stoffman, and S.E. Yonker, 2015, The people in your neighborhood: Social interactions and mutual fund portfolios, *Journal of Finance* 70: 2679-2732.
- Qi, F., C. Yang, Z. Liu, Q. Dong, M. Sun, and Z. Dong, 2019, Openhownet: An open sememe-based lexical knowledge base. arXiv preprint arXiv:1901.09957.
- Shiller, R. 1984. "Stock Prices and Social Dynamics." *Brookings Papers on Economic Activity* (2) 457-98.
- Shiller, R., and J. Pound, 1989, Survey evidence on diffusion of interest and information among investors, *Journal of Economic Behavior & Organization* 12: 47-66.
- Shue, K., 2013, Executive networks and firm policies: Evidence from the random assignment of MBA peers, *Review of Financial Studies* 26: 1401-1442.
- Tversky, A., and D. Kahneman, 1974, Judgement under uncertainty: heuristics and biases, *Science* 4157: 1124-1131.

Table 1. Summary of the investor network and their postings

We group investors into 19 groups by the number of followers. We report the number of investors in each group, the number of investors in that group as a percentage of the total number of investors, and the cumulative percentage of investors. We report the same statistics for the number of followed-follower linkages and the number of postings for each group.

Group	Number of followers	Number of investors			Number of linkages			Number of postings		
		Number	Percentage	Cumulative percentage	Number	Percentage	Cumulative percentage	Number	Percentage	Cumulative percentage
1	0	5,763	58.18%	58.18%	0	0.00%	0.00%	21985	22.86%	22.86%
2	1	2,029	20.48%	78.66%	2,029	4.99%	4.99%	14820	15.41%	38.27%
3	2	697	7.04%	85.70%	1,394	3.43%	8.43%	8415	8.75%	47.02%
4	3	341	3.44%	89.14%	1,023	2.52%	10.94%	5257	5.47%	52.49%
5	4	223	2.25%	91.39%	892	2.20%	13.14%	5621	5.84%	58.33%
6	5	140	1.41%	92.80%	700	1.72%	14.86%	3682	3.83%	62.16%
7	6	104	1.05%	93.85%	624	1.54%	16.40%	3055	3.18%	65.34%
8	7	66	0.67%	94.52%	462	1.14%	17.53%	1681	1.75%	67.08%
9	8	64	0.65%	95.16%	512	1.26%	18.79%	2934	3.05%	70.13%
10	9	41	0.41%	95.58%	369	0.91%	19.70%	1026	1.07%	71.20%
11	10	33	0.33%	95.91%	330	0.81%	20.51%	1132	1.18%	72.38%
12	11–15	117	1.18%	97.09%	1,492	3.67%	24.19%	7331	7.62%	80.00%
13	16–20	62	0.63%	97.72%	1,107	2.72%	26.91%	4255	4.42%	84.42%
14	21–30	68	0.69%	98.41%	1,661	4.09%	31.00%	3280	3.41%	87.84%
15	31–50	53	0.54%	98.94%	2,113	5.20%	36.20%	2896	3.01%	90.85%
16	51–100	48	0.48%	99.42%	3,316	8.16%	44.36%	4189	4.36%	95.20%
17	100–200	33	0.33%	99.76%	4,728	11.64%	56.00%	2833	2.95%	98.15%
18	200–1000	19	0.19%	99.95%	6,995	17.22%	73.22%	1611	1.68%	99.82%
19	>1000	5	0.05%	100%	10,882	26.78%	100%	170	0.18%	100.00%
Total		9,906	100%		40,629	100%		96,173	100.00%	

Table 2. Summary statistics

This table presents the summary statistics of the investors. *Number of followers* is the number of followers an investor has. *Number of stocks* is the number of stocks in an investor's portfolio. We first calculate the number of stocks at the individual investor level by averaging across all the days an investor has any holdings and then calculating the statistics across investors. *Number of trades per year* is the number of trades (buy and sell) during the sample period divided by the length of an investor's active period (defined as the period with at least one stock holding). *Portfolio turnover* is the annual portfolio turnover. *Number of posts per year* is the number of posts during the sample period divided by the length of an investor's active period. We separately report the number of posts per year for posts on stocks held and stocks not held. *Time owned* is the number of days since purchase. *Purchase price* is the weighted-average purchase price. *Volatility* and *Skewness* are calculated using daily returns over the six months prior to the purchase. *Volatility* is annualized. For these stock position characteristics, we first calculate the average at the individual investor level by averaging across all the purchases an investor made during our sample period and then calculating the statistics across investors. In Panel A, we report the mean, STD (standard deviation), P10 (10th percentile), P25 (25th percentile), median, P75 (75th percentile), and P90 (90th percentile) across all investors. In Panel B, we report the correlation coefficients of the investor characteristics.

Panel A. Summary statistics

	Mean	STD	P10	P25	Median	P75	P90
Number of followers	4.101	58.350	0.000	0.000	0.000	1.000	4.000
Number of stocks	3.804	3.548	1.211	1.704	2.755	4.650	7.608
Number of trades per year	105.309	151.041	16.591	33.757	66.364	121.667	211.136
Portfolio turnover	12.865	24.746	1.134	2.096	4.825	12.545	30.822
Number of posts per year	20.592	83.041	0.000	0.000	1.657	11.788	42.115
On stocks held	5.586	24.681	0.000	0.000	0.000	2.226	10.485
On stocks not held	15.006	66.626	0.000	0.000	1.029	7.799	30.737
Sqrt(Time owned)	7.209	3.866	2.041	3.913	7.143	10.033	12.426
Log(Purchase price)	2.752	0.540	2.086	2.394	2.746	3.103	3.432
Volatility	0.393	0.103	0.272	0.322	0.383	0.454	0.524
Skewness	0.294	0.321	-0.035	0.114	0.265	0.446	0.663

Panel B. Correlations

	1	2	3	4	5	6	7	8	9	10	11
1 Number of followers	1.000										
2 Number of stocks	0.041 (0.00)	1.000									
3 Number of trades per year	-0.011 (0.29)	0.141 (0.00)	1.000								
4 Portfolio turnover	-0.016 (0.11)	-0.178 (0.00)	0.610 (0.00)	1.000							
5 Number of posts per year	0.066 (0.00)	-0.017 (0.10)	0.082 (0.00)	0.108 (0.00)	1.000						
6 On stocks held	0.051 (0.00)	0.016 (0.11)	0.100 (0.00)	0.088 (0.00)	0.748 (0.00)	1.000					
7 On stocks not held	0.063 (0.00)	-0.027 (0.01)	0.065 (0.00)	0.102 (0.00)	0.969 (0.00)	0.562 (0.00)	1.000				
8 Sqrt(Time owned)	0.025 (0.01)	0.284 (0.00)	-0.374 (0.00)	-0.482 (0.00)	-0.089 (0.00)	-0.052 (0.00)	-0.091 (0.00)	1.000			
9 Log(Purchase price)	0.015 (0.14)	0.038 (0.00)	0.044 (0.00)	-0.008 (0.45)	-0.008 (0.44)	0.006 (0.57)	-0.012 (0.24)	0.033 (0.00)	1.000		
10 Volatility	-0.023 (0.02)	-0.122 (0.00)	0.270 (0.00)	0.372 (0.00)	0.044 (0.00)	0.039 (0.00)	0.041 (0.00)	-0.395 (0.00)	0.086 (0.00)	1.000	
11 Skewness	-0.011 (0.29)	-0.122 (0.00)	0.055 (0.00)	0.149 (0.00)	0.008 (0.43)	-0.017 (0.08)	0.016 (0.10)	-0.143 (0.00)	0.122 (0.00)	0.249 (0.00)	1.000

Table 3. Holding-period returns (HPR) and postings

This table presents the results of the linear probability regressions of investor posts on their holding-period returns:

$$Posting_{i,s,t+1} = \alpha + \beta_1 HPR_{i,s,t} + \gamma' X_{i,s,t} + \varepsilon_{i,s,t+1}$$

The dependent variable $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day $t+1$, and 0 otherwise. $HPR_{i,s,t}$ is the return since purchase for investor i 's holding of stock s on day t . $X_{i,s,t}$ is a set of control variables. $Sqrt(Time\ owned)$ is the square root of the number of days since purchase. $Log(Purchase\ price)$ is the natural logarithm of the weighted-average purchase price. $Volatility$ is the stock volatility calculated using daily returns over the six months prior to purchase. $Portfolio\ return$ is the total unrealized returns across all stocks held by an investor. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)
HPR	0.334*** (4.66)	0.443*** (7.07)	0.381*** (7.34)	0.128*** (3.03)	0.234*** (4.45)
Sqrt(Time owned)		-0.029*** (-9.98)	-0.003 (-1.56)	0.004 (1.60)	-0.003 (-1.22)
Log(Purchase price)		0.040 (1.53)	0.178*** (4.03)	0.64** (2.17)	0.214*** (4.23)
Volatility		0.677*** (5.16)	0.113 (1.53)	0.098* (1.68)	0.071 (0.87)
Portfolio return					0.641*** (5.51)
Constant	0.505*** (16.43)	0.421*** (4.91)			
Investor FE	NO	NO	YES	NO	YES
Date FE	NO	NO	YES	NO	YES
Stock FE	NO	NO	YES	YES	YES
Investor-Date FE	NO	NO	NO	YES	NO
Adjusted R ²	0.000	0.001	0.106	0.392	0.106
N	10,332,678	10,183,691	10,183,626	8,932,196	8,535,928

Table 4. Postings and follow-up purchases

This table presents the results on how investor postings affect followers' purchases. The unit of analysis is investor i –follower j –stock s –day t . For any day t when investor i posts on stock s (which must be in her portfolio at the end of day t), we examine how such a post affects follower j 's decision to purchase that stock relative to other stocks in investor i 's portfolio. Investor-days without posts are not included in our sample. The dependent variable $Purchase_{ij,s,t}$ is a dummy equal to 1 if follower j buys stock s in the five trading days following a post of investor i , and 0 otherwise. $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day t , and 0 otherwise. $Sqrt(Time\ owned)$ is the square root of the number of days since purchase. $Log(Purchase\ price)$ is the natural logarithm of investor i 's weighted-average purchase price. $Volatility$ is the stock volatility calculated using daily returns over the six months prior to purchase. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)
Posting	0.356*** (3.98)	0.363*** (4.16)	0.197*** (2.97)	0.198*** (2.84)
Sqrt(Time owned)		0.011** (2.04)	0.006 (0.70)	0.007 (0.81)
Log(Purchase price)		0.192 (1.43)	0.184** (2.20)	0.144 (1.11)
Volatility		-0.512 (-1.14)	-0.500 (-1.35)	-0.443 (-0.99)
Constant	0.414*** (3.95)	-0.069 (-0.20)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.001	0.001	0.014	0.021
N	2,736,886	2,721,677	2,721,394	2,718,718

Table 5. Postings and follow-up sales

This table presents the results on how investor postings affect followers' sales. The unit of analysis is investor i –follower j –stock s –day t . For any day t when investor i posts on stock s (which must be in her portfolio at the end of day t), we examine how such a posting affects follower j 's decision to sell that stock relative to other stocks in follower j 's portfolio. We only include followers who hold stock s on day t . Investor-days without postings are not included in our sample. The dependent variable $Sell_{i,j,s,t}$ is a dummy equal to 1 if follower j sells stock s in the five trading days following a post of his/her followed investor i , and 0 otherwise. $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day t , and 0 otherwise. $Sqrt(\text{Time owned})$ is the square root of the number of days since purchase. $\text{Log}(\text{Purchase price})$ is the natural logarithm of the investor's weighted-average purchase price. Volatility is the stock volatility calculated using daily returns over the six months prior to purchase. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)
Posting	2.162** (2.47)	1.719** (2.29)	0.323 (1.22)	0.328 (0.78)
Sqrt(Time owned)		-0.307*** (-4.07)	-0.102 (-1.51)	-0.023 (-0.33)
Log(Purchase price)		-0.887* (-1.67)	2.706* (1.75)	-0.934 (-0.61)
Volatility		13.387*** (3.02)	-5.291* (-1.74)	-1.175 (-0.27)
Constant	10.592*** (17.19)	12.344*** (6.65)		
Follower FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Follower-Date FE	NO	NO	NO	YES
Adjusted R ²	0.001	0.007	0.150	0.067
N	63,394	63,353	62,779	24,036

Table 6. Postings and follow-up purchases with network direction reversed

This table presents the results on how follower posts affect followed investors' purchases. The unit of analysis is follower i –followed investor j –stock s –day t . For any day t when follower i posts on stock s (which must be in her portfolio at the end of day t), we examine how such a post affects followed investor j 's decision to purchase stock s relative to other stocks in follower i 's portfolio. Follower-days without posts are not included in our sample. We also remove linkages that are bidirectional (i.e., the follower is also followed). The dependent variable $Purchase_{i,j,s,t}$ is a dummy equal to 1 if the followed investor j buys stock s in the five trading days following a post of follower i , and 0 otherwise. $Posting_{i,s,t+1}$ is a dummy equal to 1 if follower i posts about stock s on day t , and 0 otherwise. $Sqrt(Time\ owned)$ is the square root of the number of days since purchase. $Log(Purchase\ price)$ is the natural logarithm of follower i 's weighted-average purchase price. $Volatility$ is the stock volatility calculated using daily returns over the six months prior to the purchase. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)
Posting	0.036 (0.82)	0.030 (0.67)	0.018 (0.56)	0.000 (0.01)
Sqrt(Time owned)		-0.006 (-1.25)	-0.004 (-0.54)	-0.013** (-2.05)
Log(Purchase price)		0.136 (1.56)	0.020 (0.24)	-0.038 (-0.31)
Volatility		-0.373 (-1.50)	1.250** (2.06)	0.877 (1.64)
Constant	0.323*** (4.82)	0.126 (0.65)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.000	0.000	0.009	0.013
N	354,183	348,399	348,243	347,287

Table 7. Postings and follow-up purchases: Posting characteristics

This table presents the results on how the relationship between posting and follow-up purchases varies with post characteristics. We interact the *Post* dummy with post characteristics in otherwise similar regression models, as in Columns (3) and (4) of Table 4. The dependent variable $Purchase_{i,j,s,t}$ is a dummy equal to 1 if follower j buys stock s in the five trading days following a post by investor i , and 0 otherwise. $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day t , and 0 otherwise. $HPR_{i,s,t}$ is the return since purchase for investor i 's holding of stock s on day t . $PositiveSentiment_{i,s,t}$ is a dummy equal to 1 if the number of positive-sentiment words is greater than the number of negative-sentiment words, and 0 otherwise. $iOS_{i,s,t}$ is a dummy equal to 1 if the post was made using an iOS device (iPhone, iPad, or Macintosh), and 0 otherwise. *Past 6 month portfolio return* is investor i 's past six months' portfolio return. *HeldStock* is a dummy equal to 1 if the discussed stock is held by investor i , and 0 otherwise. See Table 4 for definitions of other variables. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Posting	0.201*** (3.15)	0.203*** (3.06)	0.190** (2.52)	0.226** (2.17)	0.208*** (2.75)	0.226*** (2.82)	0.157** (2.07)	0.151** (2.08)	-0.082** (-2.36)	-0.048* (-1.67)
Posting*HPR	-0.125 (-0.56)	-0.150 (-0.66)								
Posting*PositiveSentiment			0.011 (0.20)	-0.048 (-0.46)						
Posting* Past 6 months' portfolio return					0.053** (2.05)	0.081*** (4.15)				
Posting*iOS							0.272** (2.42)	0.330*** (3.07)		
Posting*HeldStock									0.279*** (4.42)	0.228*** (3.91)
HPR	0.035 (0.21)	-0.183 (-0.81)								
Past 6 months' portfolio return					-0.031 (-1.39)					
Sqrt(Time owned)	0.006 (0.71)	0.009 (0.98)	0.006 (0.71)	0.007 (0.80)	0.008 (0.95)	0.015 (1.42)	0.005 (0.65)	0.007 (0.79)		
Log(Purchase price)	0.185* (1.76)	0.079 (0.51)	0.184** (2.20)	0.143 (1.11)	0.124 (0.90)	0.062 (0.32)	0.185** (2.19)	0.134 (1.05)		
Volatility	-0.498 (-1.36)	-0.475 (-1.06)	-0.499 (-1.35)	-0.446 (-0.99)	-0.828 (-1.35)	-0.884 (-1.12)	-0.449 (-1.21)	-0.385 (-0.87)		
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Adjusted R ²	0.014	0.021	0.014	0.021	0.014	0.020	0.015	0.021	0.013	0.020
N	2,721,394	2,718,718	2,721,394	2,718,718	1,885,195	1,883,528	2,721,394	2,718,718	28,669,639	28,619,896

Table 8. Effects of stock characteristics

This table presents the results on how the relationships between holding-period return and posting (Panel A) and between posting and follow-up purchases (Panel B) vary with stock characteristics. In Panel A, we interact holding-period return (*HPR*) with a measure of stock characteristics in otherwise similar regression models, as in Columns (3) and (4) of Table 3. In Panel B, we interact *Posting* with a measure of stock characteristics in otherwise similar regression models, as in Columns (3) and (4) of Table 4. In Panel A, the dependent variable $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day $t + 1$, and 0 otherwise. $HPR_{i,s,t}$ is the return since purchase for investor i 's holding of stock s on day t . In Panel B, the dependent variable $Purchase_{i,j,s,t}$ is a dummy equal to 1 if follower j buys stock s in the five trading days following a post of investor i , and 0 otherwise. $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day t , and 0 otherwise. See Tables 3 and 4 for definitions of other variables. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A. Holding-period returns and postings

Characteristic =	Past 1 month return		Past 3 month return		Past 6 month return		Volatility		Skewness	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
HPR*Characteristic	1.466*** (4.00)	0.776*** (3.23)	0.679*** (3.66)	0.328*** (3.05)	0.281** (2.12)	0.060 (0.70)	1.303*** (4.44)	0.718*** (3.03)	0.350* (1.91)	-0.117 (-0.75)
HPR	0.334*** (6.92)	0.105*** (2.59)	0.331*** (6.91)	0.108** (2.57)	0.332*** (6.72)	0.110*** (2.66)	-0.105 (-1.03)	-0.138* (-1.78)	0.254*** (3.85)	0.145** (2.16)
Past 1 months' return	0.579*** (6.39)	0.386*** (6.12)								
Past 3 months' return			0.328*** (5.02)	0.240*** (5.63)						
Past 6 months' return					0.211*** (3.77)	0.138*** (4.67)				
Skewness									0.000 (0.00)	0.000 (0.00)
Sqrt(Time owned)	-0.004* (-1.90)	0.003 (1.24)	-0.005** (-2.21)	0.002 (1.05)	-0.006** (-2.45)	0.002 (0.81)	-0.003 (-1.42)	0.004* (1.66)	-0.004* (-1.88)	0.003 (1.41)
Log(Purchase price)	0.109*** (2.61)	0.012 (0.41)	0.086** (1.99)	-0.009 (-0.30)	0.082* (1.88)	-0.007 (-0.21)	0.147*** (3.40)	0.042 (1.45)	0.157*** (3.67)	0.044 (1.52)
Volatility	0.151** (2.04)	0.120** (1.96)	0.055 (0.79)	0.065 (1.12)	-0.064 (-0.58)	-0.015 (-0.17)	0.176** (2.23)	0.134** (2.21)	0.129* (1.76)	0.107* (1.82)
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Stock-Date FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
Adjusted R ²	0.105	0.387	0.105	0.387	0.105	0.388	0.104	0.387	0.104	0.387
N	10,372,468	9,117,362	10,368,487	9,113,552	10,298,647	9,044,317	10,372,468	9,117,362	10,372,468	9,117,362

Panel B. Postings and follow-up purchases

Characteristic =	Past 1 month return		Past 3 month return		Past 6 month return		Volatility		Skewness	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Posting*Characteristic	0.240 (1.30)	0.339 (1.26)	0.390* (1.70)	0.526** (2.10)	0.263** (2.08)	0.308** (2.52)	0.496 (1.27)	0.598 (1.37)	0.136** (2.52)	0.142** (2.18)
Posting	0.195*** (2.98)	0.194*** (2.80)	0.177*** (3.25)	0.170*** (3.11)	0.168*** (3.17)	0.162*** (2.89)	0.025 (0.17)	-0.009 (-0.05)	0.166*** (2.95)	0.165*** (2.95)
Past 1 month return	0.192 (1.04)	0.173 (0.59)								
Past 3 month return			-0.132 (-0.89)	-0.220 (-1.30)						
Past 6 month return					-0.277** (-1.97)	-0.179 (-1.11)				
Skewness									-0.049 (-1.15)	-0.053 (-1.14)
Sqrt(Time owned)	0.006 (0.68)	0.007 (0.79)	0.005 (0.65)	0.007 (0.75)	0.006 (0.75)	0.007 (0.79)	0.006 (0.71)	0.007 (0.81)	0.006 (0.68)	0.007 (0.78)
Log(Purchase price)	0.151* (1.83)	0.109 (0.86)	0.174** (1.99)	0.148 (1.08)	0.260** (2.44)	0.177 (1.17)	0.183** (2.22)	0.137 (1.08)	0.190** (2.21)	0.147 (1.15)
Volatility	-0.463 (-1.21)	-0.413 (-0.91)	-0.531 (-1.40)	-0.478 (-1.06)	-0.492 (-1.26)	-0.473 (-1.01)	-0.719 (-1.62)	-0.656 (-1.25)	-0.488 (-1.30)	-0.422 (-0.94)
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Adjusted R ²	0.015	0.021	0.015	0.021	0.015	0.021	0.015	0.021	0.015	0.021
N	2,721,394	2,718,718	2,721,244	2,718,574	2,716,299	2,713,652	2,721,394	2,718,718	2,721,394	2,718,718

Table 9. Influential investors

This table presents the results on how the relationships between holding-period return and posting (Panel A) and between posting and follow-up purchases (Panel B) vary with the number of followers an investor has. In Panel A, we interact holding-period return (*HPR*) with *Log (1 + Number of followers)* in otherwise similar regression models, as in Table 3. In Panel B, we interact *Posting* with *Log (1 + Number of followers)* in otherwise similar regression models, as in Table 4. In Panel A, the dependent variable $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor *i* posts about stock *s* on day *t* + 1, and 0 otherwise. $HPR_{i,s,t}$ is the return since purchase for investor *i*'s holding of stock *s* on day *t*. In Panel B, the dependent variable $Purchase_{i,j,s,t}$ is a dummy equal to 1 if follower *j* buys stock *s* in the five trading days following a post by investor *i*, and 0 otherwise. $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor *i* posts about stock *s* on day *t*, and 0 otherwise. See Tables 3 and 4 for definitions of other variables. All coefficients are multiplied by 100. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A. Holding-period returns and postings

	(1)	(2)	(3)	(4)
HPR*Log(1+Number of followers)	0.197 (1.44)	0.205 (1.39)	0.317*** (3.56)	0.093 (1.57)
HPR	0.048 (0.89)	0.151* (1.93)	0.137*** (2.63)	0.046 (1.29)
Log(1+Number of followers)	0.568*** (8.83)	0.573*** (8.92)		
Sqrt(Time owned)		-0.031*** (-10.56)	-0.005** (-2.10)	0.003 (1.32)
Log(Purchase price)		0.025 (1.02)	0.156*** (3.67)	0.046 (1.56)
Volatility		0.809*** (5.75)	0.129* (1.76)	0.108* (1.83)
Constant	0.159*** (6.18)	0.081 (0.87)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.006	0.007	0.105	0.387
N	10,523,380	10,372,533	10,372,468	9,117,362

Panel B. Postings and follow-up purchases

	(1)	(2)	(3)	(4)
Posting*Log(1+Number of followers)	-0.025 (-0.80)	-0.025 (-0.78)	-0.036 (-0.99)	0.016 (0.51)
Posting	0.478*** (4.14)	0.482*** (3.84)	0.378*** (2.78)	0.015 (0.14)
Log(1+ Number of followers)	-0.026* (-1.80)	-0.055** (-2.22)		
Sqrt(Time owned)		0.012** (2.20)	0.006 (0.70)	-0.014*** (-3.60)
Log(Purchase price)		0.226 (1.62)	0.179** (2.21)	0.325 (1.18)
Volatility		-0.611 (-1.35)	-0.495 (-1.34)	-0.181 (-0.20)
Constant	0.552*** (6.55)	0.145 (0.47)		
Investor FE	NO	NO	NO	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	YES	YES
Adjusted R ²	0.001	0.001	0.015	0.022
N	2,736,886	2,721,677	2,721,394	2,718,207

Table 10. Performance of follow-up purchases

This table reports the results on the size-decile portfolio adjusted buy-and-hold returns in percentage. For each purchase, we calculate the size-decile adjusted return for four windows (one, two, three, and six months) and regress it on *Follow-up purchase*. *Follow-up purchase_{j,s,t}* is a dummy equal to 1 if follower *j* bought stock *s* that any of his/her followed investors held and posted about in the past five trading days. Standard errors are clustered on three dimensions: investor, stock, and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	One month			Two months			Three months			Six months		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Follow-up purchase	-0.033 (-0.13)	0.179 (0.57)	-0.126 (-0.66)	0.375 (0.89)	0.435 (0.90)	-0.333 (-1.28)	0.611 (0.88)	0.713 (0.99)	-0.332 (-1.02)	0.731 (0.74)	1.027 (0.90)	-0.138 (-0.19)
Investor FE	YES	NO	YES	YES	NO	YES	YES	NO	YES	YES	NO	YES
Date FE	NO	YES	YES	NO	YES	YES	NO	YES	YES	NO	YES	YES
Stock FE	NO	YES	YES	NO	YES	YES	NO	YES	YES	NO	YES	YES
Adjusted R ²	0.036	0.054	0.114	0.053	0.071	0.171	0.063	0.081	0.216	0.079	0.093	0.305
N	609,479	609,478	609,459	609,479	609,478	609,459	609,479	609,478	609,459	609,479	609,478	609,459

Figure 1. Snowball user interface

This figure provides screenshots of the Snowball user interface. Panel A shows a user profile in the view of herself. Panel B shows a user profile in the view of other users. Panel C shows the trading and performance records of an investor, also in the view of other users. We also provide a translation of part of the pages.

Panel A. User's own homepage

Market indexes

Users own portfolio holdings and the daily and year-to-day performance of each holding

Postings of the investors followed by the user, ranked newest to oldest

Panel B. Profile of a followed investor

Username, number of followed users, number of followers, location, and a short self-portrait of investment style

Three postings by this investor, with the date, time, and device used

Link to brokerage account

Panel C. Trading and performance records of the followed investor

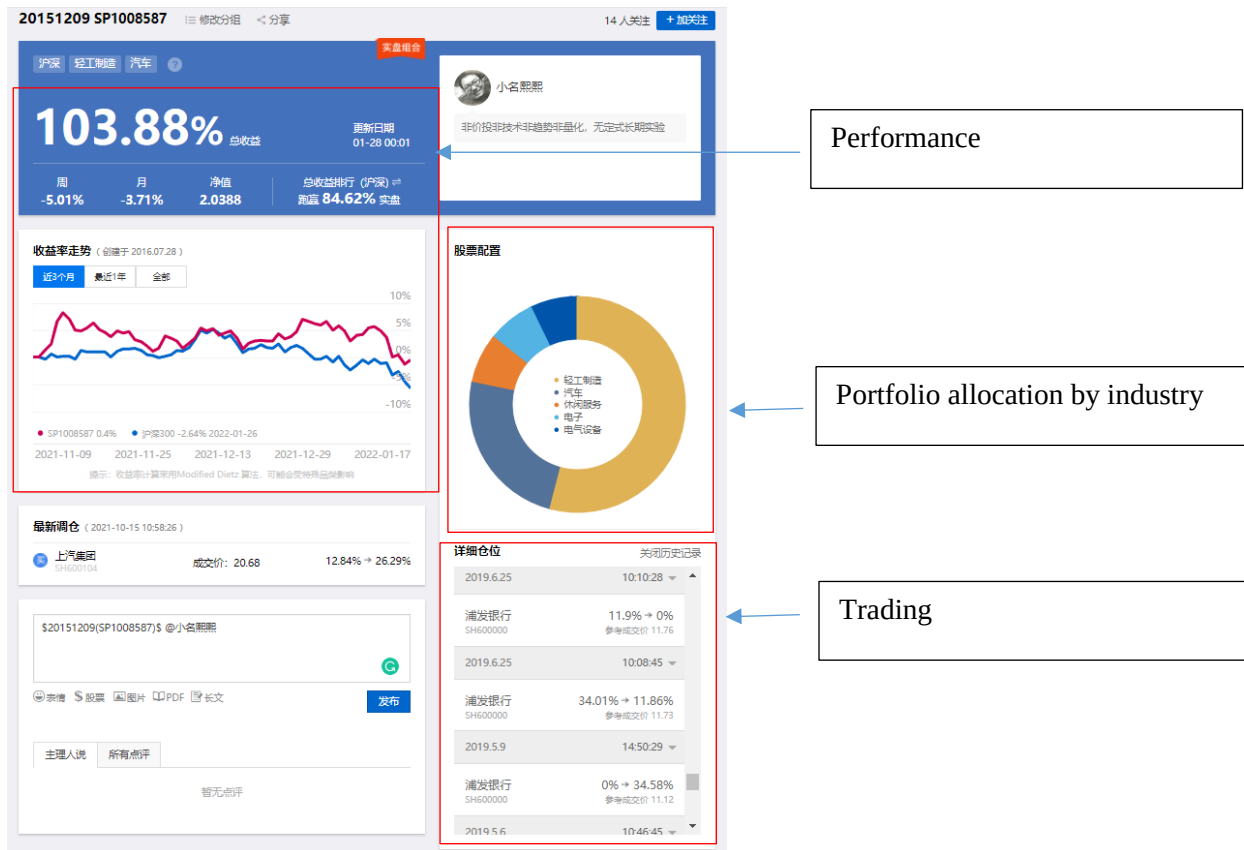


Figure 2. Distribution of Snowball investors by province

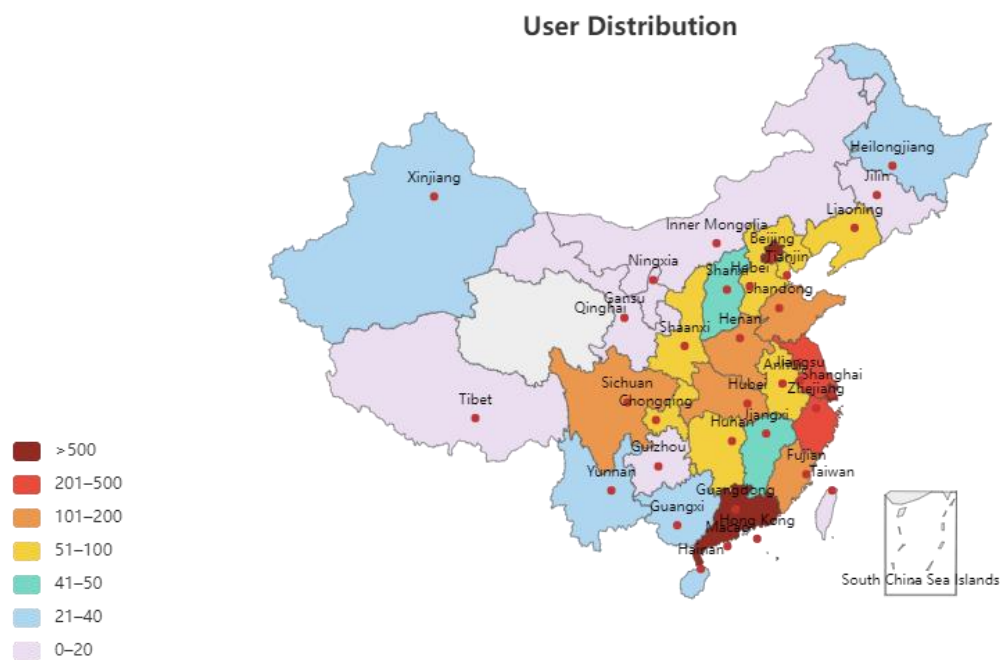


Figure 3. A graphic view of the relationship between holding-period returns and postings

This figure plots the coefficients (multiplied by 100) of the dummy variables representing a 5% interval of holding-period returns. The omitted group is $(-\infty, -40\%)$. The dummy variables replace the holding-period return variable in a linear probability regression similar to that in Column 3 of Table 2:

$$Posting_{i,s,t+1} = \alpha + \beta_1 I_{\{HPR_{i,s,t} \in [-40\%, -35\%]\}} + \beta_2 I_{\{HPR_{i,s,t} \in [-35\%, -30\%]\}} \dots + \beta_{17} I_{\{HPR_{i,s,t} \in [40\%, +\infty]\}} + \gamma' X_{i,s,t} + \varepsilon_{i,s,t+1}$$

The dependent variable $Posting_{i,s,t+1}$ is a dummy equal to 1 if investor i posts about stock s on day $t+1$, and 0 otherwise. $HPR_{i,s,t}$ is the return since purchase for investor i 's holding of stock s on day t . $X_{i,s,t}$ is a set of control variables. $Sqrt(Time\ owned)$ is the square root of the number of days since purchase. $Log(Buy\ Price)$ is the natural logarithm of the weighted-average purchase price. $Volatility$ is the stock volatility calculated using daily returns over the six months prior to purchase. The regression includes investor, stock, and day fixed effects. Standard errors are clustered on three dimensions: investor, stock, and day. The solid blue line plots the coefficients (multiplied by 100) on the dummy variables; the dotted lines show the upper and lower bounds of the 95% confidence intervals.



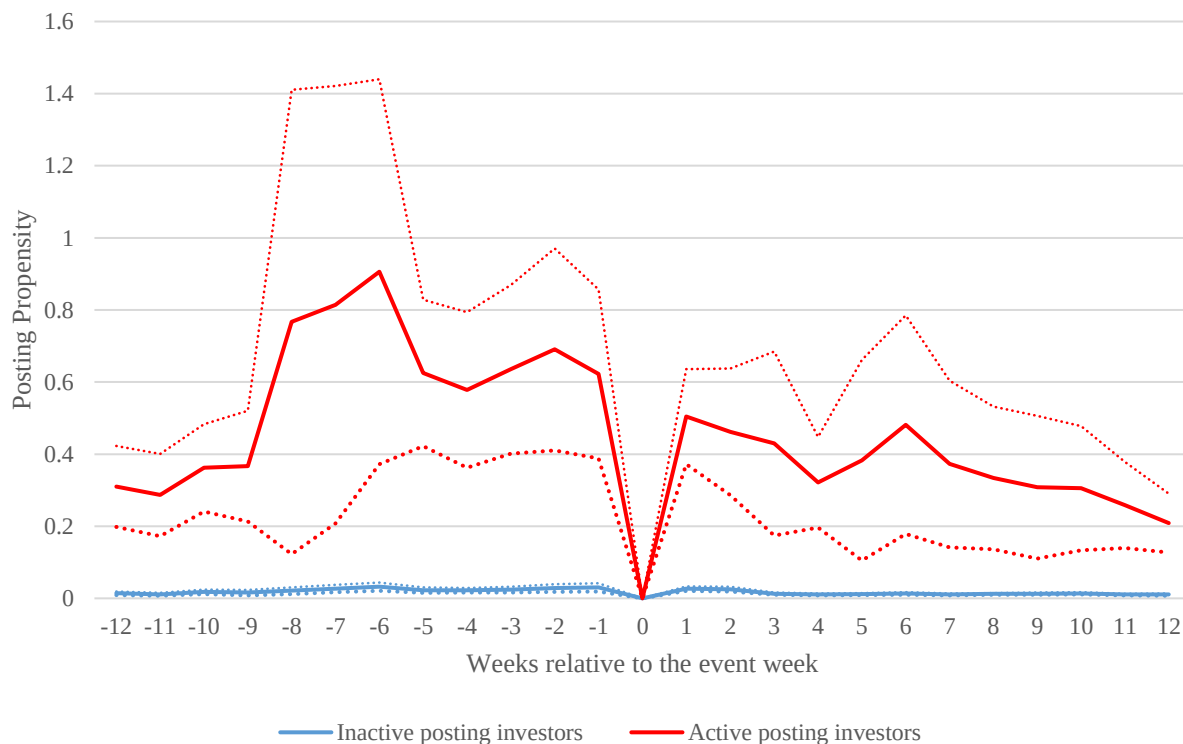
Figure 4. Effect of disabling the post function

Panel A reports the posting propensity of two groups of investors. An investor's posting propensity is the average daily probability that she will post about a held stock, calculated as the number of posts divided by the number of investor-stock-day observations, calculated for the 90-day period from January 14, 2019, to April 12, 2019. Active posting investors are those whose posting propensity is higher than 0.1. Panel B displays the β_w coefficients (multiplied by 100) from the regression:

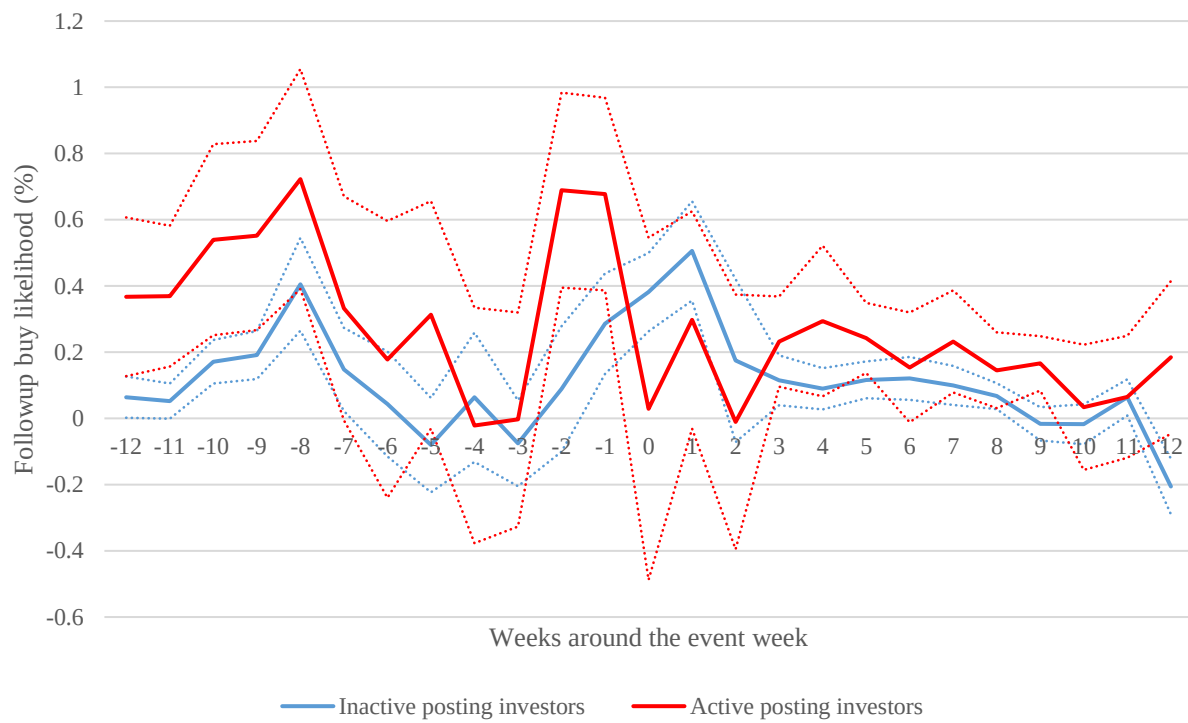
$$Purchase_{j(i),s,(t+1,t+5)} = \alpha + \sum_{w=-12}^{12} \beta_w Gain_{i,s,t} * RelativeWeek_w + \varepsilon_{i,j,s,(t+1,t+5)}$$

The dependent variable $Purchase_{j(i),s,(t+1,t+5)}$ is a dummy equal to 1 if follower j buys stock s in the five trading days following a post of his/her followed investor i , and 0 otherwise. $Gain_{i,s,t}$ is a dummy equal to 1 if investor i 's holding-period return on stock s on day t is positive, and 0 otherwise. $RelativeWeek_w$ is a dummy equal to 1 if day t is in the week that is w weeks after the week when the post function was disabled. We estimate the above equation separately for active and inactive posting investors. Standard errors are clustered on three dimensions: investor, stock, and day. The two solid lines plot the β_w coefficients (multiplied by 100); the dotted lines show the upper and lower bounds of the 95% confidence intervals.

Panel A. Posting propensity



Panel B. Follow-up purchases



Internet Appendix

Figure A1. Active and closed accounts

This figure reports the density distribution of total returns of the active and closed accounts. Closed accounts with exactly 0% return are excluded. Among 30,513 investors, there are 5,299 accounts with exactly 0% return. Many accounts are only open for a very short period; the concentration of closed accounts with around 0% return in Panel A is largely because of these very short-lived no-trading accounts. We require the duration of an account to be longer than 1 day in Panel A, 180 days in Panel B, and 365 days in Panel C.

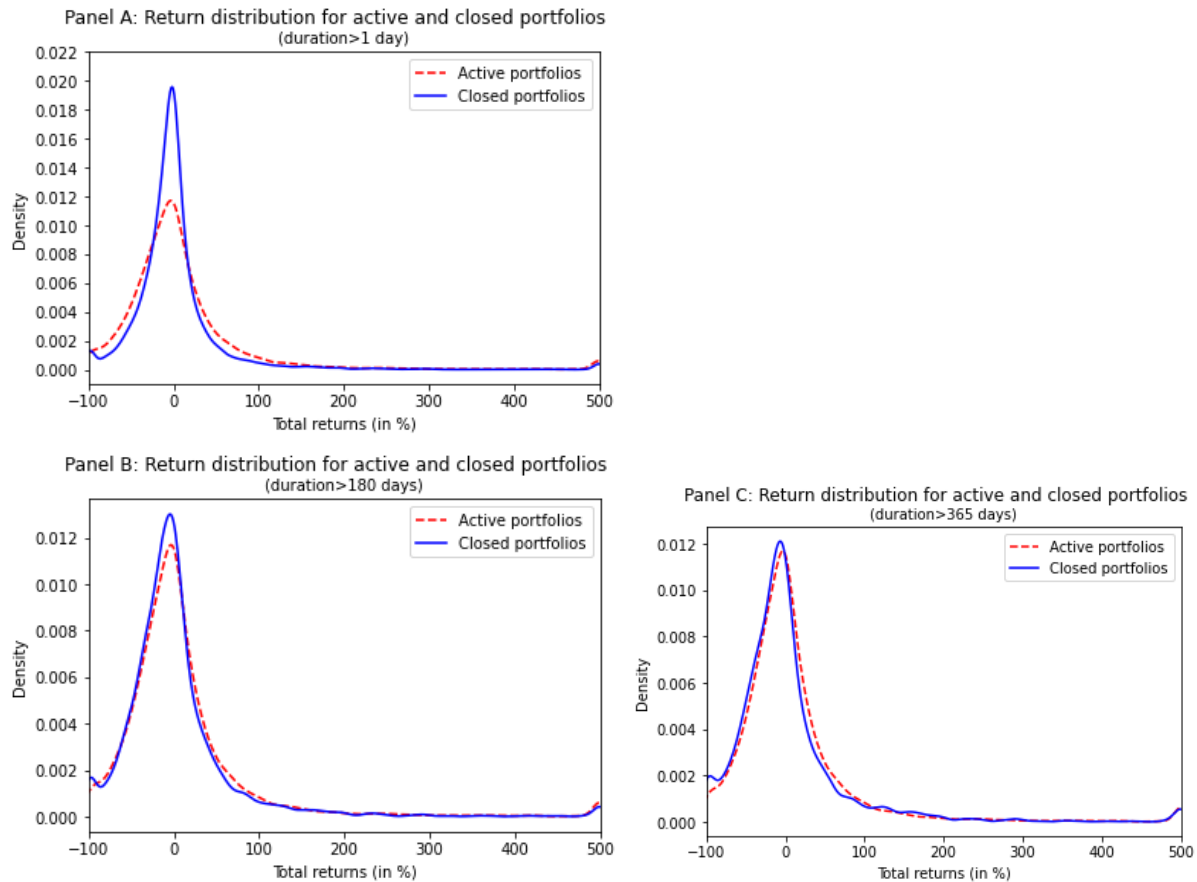


Table A1. Total portfolio returns and account closing

This table reports the results on how portfolio returns are associated with whether an account is closed or not. We construct our sample as follows: For each day t on which at least one account is closed, we keep these closed accounts and all the active accounts that were open before that day. The dependent variable is a dummy variable equal to 1 if an account is closed, and 0 otherwise. *Total return* is the total return of a portfolio since its inception. We set *Total return* to -99% if it is lower than -99%, and to 500% if it is higher than 500%. *Sqrt(Duration)* is the square root of the number of calendar days since an account's opening. *Opening date* is the account opening date. *Date* is analogous to day t in the above example. All coefficients are multiplied by 10,000. Standard errors are clustered by date and opening date. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Full sample			Duration>180		
	(1)	(2)	(3)	(4)	(5)	(6)
Total return	0.040 (1.35)	0.097** (2.41)	0.051* (1.91)	0.014 (0.88)	0.024 (1.40)	0.022 (1.32)
Sqrt(Duration)		-4.684*** (-19.27)			-1.157*** (-9.17)	
Date FE	YES	YES	NO	YES	YES	NO
Opening date*Date FE	NO	NO	YES	NO	NO	YES
Adjusted R ²	0.000	0.004	0.149	0.001	0.001	0.023
N	7,143,972	7,143,972	7,141,087	5,585,142	5,585,142	5,583,940

Table A2. Holding-period returns and postings excluding investors without followers

This table is the same as Table 3, except that we exclude investors without any followers from the analysis.

	(1)	(2)	(3)	(4)	(5)
HPR	0.435*** (3.48)	0.651*** (5.68)	0.520*** (5.33)	0.161** (2.01)	0.302*** (3.06)
Sqrt(Time owned)		-0.049*** (-8.22)	-0.005 (-1.11)	0.004 (0.82)	-0.004 (-0.97)
Log(Purchase price)		0.089 (1.59)	0.281*** (3.01)	0.099 (1.59)	0.323*** (3.06)
Volatility		1.179*** (3.58)	0.051 (0.32)	0.089 (0.78)	-0.012 (-0.07)
Portfolio return					0.973*** (4.39)
Constant	0.859*** (14.15)	0.662*** (3.56)			
Investor FE	NO	NO	YES	NO	YES
Date FE	NO	NO	YES	NO	YES
Stock FE	NO	NO	YES	YES	YES
Investor-Date FE	NO	NO	NO	YES	NO
Stock-Date FE	NO	NO	NO	NO	NO
Adjusted R ²	0.000	0.002	0.121	0.411	0.122
N	4,607,235	4,543,139	4,543,096	3,999,368	3,805,188

Table A3. Postings and follow-up purchases: Next one or three trading days

Panel A. Follow-up purchases: Next one trading day

	(1)	(2)	(3)	(4)
Posting	0.146*** (2.82)	0.148*** (2.82)	0.102* (1.78)	0.112* (1.77)
Sqrt(Time owned)		0.003 (1.39)	-0.001 (-0.22)	-0.001 (-0.12)
Log(Purchase price)		0.081 (1.40)	0.081*** (2.88)	0.051 (0.96)
Volatility		-0.193 (-1.03)	-0.325* (-1.86)	-0.280 (-1.18)
Constant	0.160*** (3.71)	-0.032 (-0.22)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.000	0.000	0.008	0.016
N	2,736,886	2,721,677	2,721,394	2,718,718

Panel B. Follow-up purchases: Next three trading days

	(1)	(2)	(3)	(4)
Posting	0.270*** (3.68)	0.274*** (3.78)	0.165** (2.56)	0.170** (2.46)
Sqrt(Time owned)		0.007* (1.90)	0.003 (0.43)	0.004 (0.52)
Log(Purchase price)		0.138 (1.42)	0.152*** (2.84)	0.119 (1.36)
Volatility		-0.326 (-0.99)	-0.387 (-1.37)	-0.390 (-1.11)
Constant	0.291*** (3.89)	-0.062 (-0.24)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.000	0.001	0.012	0.019
N	2,736,886	2,721,677	2,721,394	2,718,718

Table A4. Postings and follow-up purchases using network data measured at a different date

This table presents the results on how investor posts affect followers' purchases. The specifications are the same as in Table 3, except that we use the network snapshot collected in October 2020 instead of May 2019.

	(1)	(2)	(3)	(4)
Posting	0.355*** (3.78)	0.358*** (4.02)	0.181*** (2.98)	0.187*** (2.86)
Sqrt(Time owned)		0.013** (2.41)	0.002 (0.21)	-0.001 (-0.12)
Log(Purchase price)		0.204 (1.50)	0.060 (0.64)	0.013 (0.10)
Volatility		-0.614 (-1.28)	-0.553 (-1.57)	-0.486 (-1.01)
Constant	0.446*** (4.11)	-0.054 (-0.16)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.001	0.001	0.014	0.021
N	2,724,504	2,710,605	2,710,303	2,707,858

Table A5. Impact of truncation

This table presents the results on how investor postings affect followers' purchases. The specifications are the same as in Table 4, except that we deal with the truncation in the maximum number of trades in our data. Specifically, we remove the follower-days on which the follower's trades are truncated. For example, if follower j 's trades are truncated and her earliest displayed trade was on day t , we only include her in the analysis after day t .

	(1)	(2)	(3)	(4)
Posting	0.500*** (4.17)	0.509*** (4.49)	0.237*** (2.97)	0.244*** (2.82)
Sqrt(Time owned)		-0.000 (-0.09)	0.008 (0.72)	0.009 (0.85)
Log(Purchase price)		0.252 (1.52)	-0.091 (-0.63)	-0.010 (-0.06)
Volatility		-1.142* (-1.84)	-0.513 (-1.04)	-0.377 (-0.60)
Constant	0.504*** (3.87)	0.167 (0.42)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.001	0.002	0.020	0.029
N	2,128,276	2,118,354	2,118,056	2,115,436

Table A6. Postings and follow-up purchases at the investor-stock-day level

This table presents the results on how investor postings affect followers' purchases. The specifications are the same as in Table 4, except that the unit of analysis is investor-stock-day instead of investor-follower-stock-day. In Panel A, the dependent variable is a dummy equal to 1 if any of investor *i*'s followers buy stock *s* in the five trading days following a post by this investor, and 0 otherwise. In Panel B, the dependent variable is the fraction of investor *i*'s followers who buy stock *s* in the five trading days following a post by this investor.

Panel A. Dependent variable is the fraction of followers who buy

	(1)	(2)	(3)	(4)
Posting	0.478*** (5.72)	0.475*** (6.02)	0.317*** (5.31)	0.341*** (4.88)
Sqrt(Time owned)		-0.000 (-0.04)	-0.008 (-0.86)	-0.002 (-0.18)
Log(Purchase price)		0.247* (1.84)	0.209 (1.08)	0.252 (1.07)
Volatility		-0.466 (-1.51)	-0.007 (-0.01)	-0.421 (-0.63)
Constant	0.559*** (6.14)	0.046 (0.13)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.002	0.003	0.119	0.124
N	107,493	105,567	104,970	95,747

Panel B. Dependent variable is a dummy indicating if any follower buys

	(1)	(2)	(3)	(4)
Posting	3.898*** (5.12)	4.127*** (5.52)	2.364*** (6.82)	2.241*** (5.79)
Sqrt(Time owned)		0.193** (2.08)	0.048 (0.82)	0.048 (0.69)
Log(Purchase price)		2.313* (1.92)	3.814*** (2.80)	3.763*** (2.73)
Volatility		-5.208* (-1.73)	-3.365 (-1.05)	-4.762 (-1.35)
Constant	5.259*** (5.65)	-0.846 (-0.27)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.006	0.015	0.295	0.314
N	107,493	105,567	104,970	95,747

Table A7. Past market returns

In Panel A, we examine how the relationship between holding-period returns and posting is related to past market returns. In Panel B, we examine how the relationship between postings and follow-up purchases is related to past market returns. We consider four horizons of past market returns, ranging from the past three months to the past two years. See Tables 3 and 4 for definitions of variables.

Panel A. Holding-period returns and postings

Past market returns	Past 3 months		Past 6 months		Past 1 year		Past 2 years	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HPR* Past market returns	0.380 (0.69)	0.168 (0.42)	0.115 (0.30)	-0.026 (-0.09)	-0.169 (-0.64)	-0.118 (-0.68)	0.090 (0.71)	-0.015 (-0.13)
HPR	0.358*** (6.95)	0.117*** (2.70)	0.366*** (6.84)	0.118*** (2.72)	0.371*** (7.22)	0.112*** (2.68)	0.349*** (6.60)	0.113*** (2.72)
Past market returns	0.137 (0.96)	0.189* (1.65)	0.158 (1.49)	0.114 (1.43)	0.160* (1.73)	-0.008 (-0.10)	-0.036 (-0.58)	0.015 (0.33)
Sqrt(Time owned)	-0.004* (-1.82)	0.003 (1.43)	-0.004* (-1.90)	0.003 (1.33)	-0.004** (-1.97)	0.003 (1.35)	-0.004 (-1.49)	0.003 (1.19)
Log(Purchase price)	0.151*** (3.54)	0.042 (1.42)	0.148*** (3.50)	0.042 (1.41)	0.146*** (3.40)	0.045 (1.54)	0.157*** (3.67)	0.045 (1.52)
Volatility	0.135* (1.81)	0.116* (1.93)	0.142* (1.92)	0.117* (1.95)	0.163** (2.09)	0.105* (1.73)	0.132* (1.81)	0.106* (1.81)
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES
Adjusted R ²	0.104	0.387	0.104	0.387	0.104	0.387	0.104	0.387
N	10,372,468	9,117,362	10,372,468	9,117,362	10,372,468	9,117,362	10,372,468	9,117,362

Panel B. Postings and follow-up purchases

Past market returns	Past 3 months		Past 6 months		Past 1 year		Past 2 years	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Posting*Past market returns	-0.490 (-1.56)	-0.199 (-0.62)	-0.232 (-0.61)	-0.038 (-0.11)	0.095 (0.28)	0.226 (0.70)	0.061 (0.22)	0.238 (0.73)
Posting	0.193*** (2.98)	0.196*** (2.85)	0.196*** (2.92)	0.197*** (2.80)	0.199*** (2.94)	0.201*** (2.86)	0.195*** (3.23)	0.189*** (3.11)
Past market returns	0.299 (0.83)	0.256 (0.72)	-0.088 (-0.25)	0.063 (0.15)	0.320 (1.23)	0.382 (1.04)	-0.035 (-0.16)	0.029 (0.12)
Sqrt(Time owned)	0.006 (0.73)	0.007 (0.83)	0.006 (0.69)	0.007 (0.82)	0.006 (0.73)	0.008 (0.88)	0.006 (0.65)	0.006 (0.61)
Log(Purchase price)	0.181** (2.30)	0.135 (1.12)	0.195** (2.43)	0.141 (1.12)	0.160* (1.79)	0.118 (0.86)	0.186** (2.24)	0.148 (1.14)
Volatility	-0.486 (-1.34)	-0.422 (-0.98)	-0.554 (-1.60)	-0.430 (-1.00)	-0.328 (-0.81)	-0.234 (-0.44)	-0.498 (-1.34)	-0.429 (-0.96)
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES
Stock-Date FE	NO	NO	NO	NO	NO	NO	NO	NO
Adjusted R ²	0.014	0.021	0.014	0.021	0.014	0.021	0.014	0.021
N	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718

Table A8. Postings and follow-up purchases: Sentiment

This table reports the results of various ways of defining the sentiment of postings. In Panel A, we define the sentiment based on the word lists we developed in the financial contexts. In Panel B, we define the sentiment based on HowNet. This table is the same as Columns (3) and (4) of Table 7, except we measure sentiment in different ways.

Panel A. Based on word lists developed by the authors

Sentiment =	PositiveSentiment dummy		NegativeSentiment dummy		Positive/Total		Negative/Total		(Positive-Negative) /Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Posting	0.190** (2.52)	0.226** (2.17)	0.196*** (3.04)	0.186*** (2.92)	0.184*** (2.66)	0.218** (2.27)	0.193*** (2.91)	0.182*** (2.74)	0.195*** (2.92)	0.215*** (2.74)
Posting*Sentiment	0.011 (0.20)	-0.048 (-0.46)	0.015 (0.20)	0.107 (1.04)	0.021 (0.37)	-0.034 (-0.36)	0.027 (0.54)	0.094* (1.77)	0.004 (0.11)	-0.039 (-0.70)
Sqrt(Time owned)	0.006 (0.71)	0.007 (0.80)	0.006 (0.70)	0.007 (0.81)	0.006 (0.72)	0.007 (0.81)	0.006 (0.70)	0.007 (0.81)	0.006 (0.71)	0.007 (0.80)
Log(Purchase price)	0.184** (2.20)	0.143 (1.11)	0.184** (2.20)	0.143 (1.11)	0.185** (2.21)	0.143 (1.11)	0.185** (2.21)	0.143 (1.11)	0.184** (2.20)	0.143 (1.10)
Volatility	-0.499 (-1.35)	-0.446 (-0.99)	-0.499 (-1.35)	-0.440 (-0.98)	-0.500 (-1.35)	-0.444 (-0.99)	-0.499 (-1.34)	-0.439 (-0.98)	-0.500 (-1.35)	-0.442 (-0.98)
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Adjusted R ²	0.014	0.021	0.014	0.021	0.014	0.021	0.014	0.021	0.014	0.021
N	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718

Panel A. Based on word lists developed by HowNet

Sentiment =	PositiveSentiment dummy		NegativeSentiment dummy		Positive/Total		Negative/Total		(Positive-Negative) /Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Posting	0.215*** (3.60)	0.225*** (4.03)	0.197*** (2.95)	0.198*** (2.80)	0.233*** (3.40)	0.240*** (3.50)	0.188*** (2.89)	0.183*** (2.63)	0.220*** (3.25)	0.227*** (3.32)
Posting*Sentiment	-0.022 (-0.69)	-0.035 (-0.83)	0.010 (0.19)	-0.015 (-0.21)	-0.046 (-1.05)	-0.054 (-1.03)	0.069 (1.27)	0.104 (1.54)	-0.035 (-1.28)	-0.046 (-1.34)
Sqrt(Time owned)	0.006 (0.71)	0.007 (0.82)	0.006 (0.70)	0.007 (0.81)	0.006 (0.71)	0.007 (0.82)	0.006 (0.70)	0.007 (0.80)	0.006 (0.71)	0.007 (0.82)
Log(Purchase price)	0.184** (2.20)	0.144 (1.11)	0.184** (2.20)	0.143 (1.11)	0.183** (2.19)	0.143 (1.11)	0.183** (2.18)	0.143 (1.10)	0.183** (2.19)	0.143 (1.10)
Volatility	-0.500 (-1.35)	-0.443 (-0.99)	-0.500 (-1.35)	-0.443 (-0.99)	-0.500 (-1.35)	-0.443 (-0.99)	-0.498 (-1.34)	-0.438 (-0.98)	-0.499 (-1.35)	-0.441 (-0.98)
Investor FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Date FE	YES	NO	YES	NO	YES	NO	YES	NO	YES	NO
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Investor-Date FE	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Adjusted R ²	0.014	0.021	0.014	0.021	0.014	0.021	0.014	0.021	0.014	0.021
N	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718	2,721,394	2,718,718

Table A9. Postings and follow-up purchases: Past-one-year portfolio return

This table is the same as Columns (5) and (6) of Table 7, except we measure the past portfolio returns over a year. *Past 1 year portfolio return* is subsumed by the date fixed effects (Column 1) or by the investor-day fixed effects (Column 2).

	(1)	(2)
Posting*Past 1 year portfolio return	0.043** (2.58)	0.057*** (4.02)
Posting	0.227*** (2.69)	0.241*** (2.78)
Past 1 year portfolio return	-0.002 (-0.07)	
Sqrt(Time owned)	0.004 (0.54)	0.005 (0.57)
Log(Purchase price)	-0.213 (-1.39)	-0.323 (-1.34)
Volatility	-1.035 (-1.47)	-0.990 (-1.13)
Investor FE	YES	NO
Date FE	YES	NO
Stock FE	YES	YES
Investor-Date FE	NO	YES
Adjusted R ²	0.015	0.020
N	1457900	1456829

Table A10. Influential investors: Considering followers without a linked brokerage account

This table is the same as Table 9, except we define the number of followers by including those without a linked brokerage account.

Panel A. Holding-period returns and postings

	(1)	(2)	(3)	(4)
HPR*Log(1+Number of followers)	0.175 (1.54)	0.189 (1.54)	0.298*** (3.94)	0.084 (1.64)
HPR	0.019 (0.35)	0.114 (1.39)	0.095* (1.76)	0.037 (0.99)
Log(1+Number of followers)	0.509*** (9.25)	0.512*** (9.33)		
Sqrt(Time owned)		-0.030*** (-10.52)	-0.005** (-2.12)	0.003 (1.31)
Log(Purchase price)		0.024 (0.97)	0.155*** (3.66)	0.046 (1.56)
Volatility		0.804*** (5.71)	0.131* (1.78)	0.108* (1.84)
Constant	0.109*** (4.03)	0.035 (0.37)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.006	0.007	0.105	0.387
N	10,523,380	10,372,533	10,372,468	9,117,362

Panel B. Postings and follow-up purchases

	(1)	(2)	(3)	(4)
Posting*Log(1+Number of followers)	-0.025 (-0.83)	-0.026 (-0.80)	-0.036 (-1.01)	0.015 (0.48)
Posting	0.494*** (3.85)	0.497*** (3.58)	0.397*** (2.64)	0.013 (0.11)
Log(1+Number of followers)	-0.026* (-1.83)	-0.053** (-2.21)		
Sqrt(Time owned)		0.012** (2.20)	0.006 (0.70)	-0.014*** (-3.61)
Log(Purchase price)		0.225 (1.62)	0.178** (2.21)	0.324 (1.18)
Volatility		-0.606 (-1.34)	-0.495 (-1.34)	-0.181 (-0.20)
Constant	0.565*** (6.59)	0.166 (0.54)		
Investor FE	NO	NO	NO	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	YES	YES
Adjusted R ²	0.001	0.001	0.015	0.022
N	2,736,886	2,721,677	2,721,394	2,718,207

Table A11. Influential investors: Alternate specification

This table is the same as Table 9, except we relax the linearity assumption. Specifically, we group investors based on the number of followers and then interact group dummies with either *HPR* (Panel A) or *Posting* (Panel B). The base group is the investors with more than 100 followers.

Panel A. Holding-period returns (HPR) and postings

	(1)	(2)	(3)	(4)
HPR	1.017 (0.83)	1.080 (0.85)	1.887* (1.78)	1.113 (1.41)
HPR*I _{Number of followers=0}	-0.924 (-0.77)	-0.921 (-0.72)	-1.758* (-1.66)	-1.063 (-1.35)
HPR*I _{Number of followers∈[1,20]}	-0.674 (-0.56)	-0.582 (-0.46)	-1.317 (-1.24)	-0.934 (-1.19)
HPR*I _{Number of followers∈[21,100]}	0.049 (0.04)	0.010 (0.01)	-1.336 (-1.18)	-1.460* (-1.68)
I _{Number of followers=0}	-2.275*** (-3.11)	-2.278*** (-3.11)	0.000 (0.00)	0.000 (0.00)
I _{Number of followers∈[1,20]}	-1.743** (-2.38)	-1.734** (-2.37)		
I _{Number of followers∈[21,100]}	0.158 (0.18)	0.181 (0.21)		
Sqrt(Time owned)		-0.030*** (-10.20)	-0.005** (-2.01)	0.003 (1.34)
Log(Purchase price)		0.031 (1.26)	0.155*** (3.65)	0.046 (1.57)
Volatility		0.749*** (5.51)	0.131* (1.78)	0.110* (1.87)
Constant	2.488*** (3.40)	2.408*** (3.28)		
Investor FE	NO	NO	YES	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	NO	YES
Adjusted R ²	0.004	0.005	0.104	0.387
N	10,523,380	10,372,533	10,372,468	9,117,362

Panel B. Postings and follow-up purchases

	(1)	(2)	(3)	(4)
Posting	0.361*** (3.46)	0.366*** (3.49)	0.146 (1.53)	0.114 (1.29)
Posting* $I_{\{Number\ of\ followers \in [1,20]\}}$	0.236** (2.33)	0.243** (2.30)	0.299*** (2.61)	0.022 (0.28)
Posting* $I_{\{Number\ of\ followers \in [21,100]\}}$	-0.183** (-1.97)	-0.184* (-1.89)	-0.003 (-0.04)	-0.083 (-1.11)
$I_{\{Number\ of\ followers \in [1,20]\}}$	0.160** (2.29)	0.197** (2.01)		
$I_{\{Number\ of\ followers \in [21,100]\}}$	-0.023 (-0.25)	0.013 (0.13)		
Sqrt(Time owned)		0.011** (2.12)	0.006 (0.67)	-0.014*** (-3.71)
Log(Purchase price)		0.196 (1.48)	0.177** (2.17)	0.322 (1.16)
Volatility		-0.529 (-1.18)	-0.486 (-1.30)	-0.176 (-0.20)
Constant	0.391*** (3.22)	-0.110 (-0.32)		
Investor FE	NO	NO	NO	NO
Date FE	NO	NO	YES	NO
Stock FE	NO	NO	YES	YES
Investor-Date FE	NO	NO	YES	YES
Adjusted R ²	0.001	0.001	0.015	0.022
N	2,736,886	2,721,677	2,721,394	2,718,207